Ti-6Al-2Sn-4Zr-2Mo-0.08Si

Common Name: Ti-6242S Ti-6242Si
UNS Number: R54620

Ti-6Al-2Sn-4Zr-2Mo-0.08Si (Ti-6242S or Ti-6242Si), developed in the late 1960s as an elevated-temperature alloy, has an outstanding combination of tensile strength, creep strength, toughness, and high-temperature stability for long-term applications at temperatures up to 425 °C (800 °F). Ti-6242S is one of the most creep-resistant titanium alloys and is recommended for use up to 565 °C (1050 °F). Proper heat treatment is important in allowing the alloy to develop its maximum creep resistance.

Chemistry and Density

The 6 percent aluminum addition in the Ti-6Al-2Sn-4Zr-2Mo composition is a potent alpha-phase stabilizer, while the 2 percent molybdenum addition represents only a moderate quantity of this potent beta-phase stabilizer. The tin and zirconium additions are solid-solution strengthening elements that are neutral with respect to phase stabilization. The net effect of this combination of alloying elements is the generation of a weakly beta-stabilized, alpha-beta alloy. Since it is weakly beta stabilized, the alloy is also properly described as a near-alpha, alpha-beta alloy.

The original composition of this alloy contained no silicon, but RMI introduced a nominal 0.08% silicon content which allowed the alloy to meet the creep requirements for its intended jet-engine applications. Before any major commercial applications were developed, all producers had added silicon to the original Ti-6242 composition.

Density. 4.54 g/cm^3 (0.164 lb/in.^{3})

Product Forms

Available mill forms include billet, bar, plate, sheet, strip, and extrusions. Cast Ti-6242S products constitute about 7% of cast titanium products. Some forming operations can be carried out at room temperature, and warm forming (425 to 705 °C, or 800 to 1300 °F) is employed when necessary. Ti-6242S has fair weldability. The molten weld metal and adjacent heated zones must be shielded from active gases (nitrogen, oxygen, and hydrogen).

Product Condition/ Microstructure

Ti-6242S is sometimes described as a near-alpha or superalpha alloy, but in its normal heat treated condition this alloy has a structure better described as alpha-beta. Proper treatment is needed to develop good creep resistance. Limited hardening of Ti-6242S can be done by solution treating and aging.

Applications

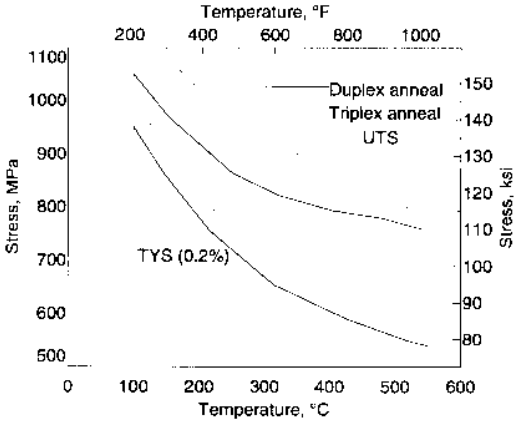
Ti-6242S is used primarily for gas turbine components such as compressor blades, disks, and impellers, and also in sheet-metal form for engine afterburner structures and for various "hot" airframe skin applications, where high strength and toughness, excellent creep resistance, and stress stability at temperatures up to 565 °C (1050 °F) are required.

Ti-6Al-2Sn-4Zr-2Mo-0.08Si: Specifications and compositions

			Composition, wt%								
Specification	Designation	Description	Al	Fe	H	Mo	N	O	Sn	Zr	Other
UNS	R54620		6			2			2	4	bal Ti
UNS	R54621	Weld Fill Met	5.5-6.5	0.05	0.015	1.8-2.2	0.15	0.3	1.8-2.2	3.6-4.4	C 0.04; Cr 0.25; bal Ti
Germany											
	WL 3.7144		5.5-6.5	0.25	0.015	1.8-2.2	0.05	0.15	1.8-2.2	3.6-4.4	C 0.05; bal Ti
Spain											
UNE 38-718	L-7103	Sh Strp Plt Ann	5.5-6.5	0.25	0.015	1.8-2.2	0.05	0.12	1.8-2.2	3.6-4.4	C 0.05; OT 0.4; bal Ti
UNE 38-718	L-7103	Sh Strp Plt HT	5.5-6.5	0.25	0.015	1.8-2.2	0.05	0.12	1.8-2.2	3.6-4.4	C 0.05; OT 0.4; bal Ti
USA											
AMS 4919C		Sh Strp Plt	5.5-6.5	0.25 max	0.015 max	1.8-2.2	0.05 max	0.12 max	1.8-2.2	3.6-4.4	C 0.05 max; Si 0.06-0.1; Y 0.005 max; OE 0.1 max; OT 0.3 max; bal Ti
AMS 4919G		Sh Strp Plt DA	5.5-6.5	0.25	0.015	1.8-2.2	0.05	0.12	1.8-2.2	3.6-4.4	C 0.05; Si 0.1; Y 0.005; OT 0.3; bal Ti
AMS 4975E		Bar Wir Rng Bil STA	5.5-6.5	0.25	0.0125	1.8-2.2	0.05	0.15	1.8-2.2	3.6-4.4	C 0.05; Y 0.005; OT 0.3; Si 0.1; bal Ti
AMS 4975F		Bar Rng HT	5.5-6.5	0.1 max	0.0125 max	1.8-2.2	0.05 max	0.15 max	1.8-2.2	3.6-4.4	C 0.05 max; Si 0.06-0.1; Y 0.005 max; OE 0.1 max; OT 0.3 max; bal Ti
AMS 4976C		Frg STA	5.5-6.5	0.25	0.0125	1.8-2.2	0.05	0.15	1.8-2.2	3.6-4.4	C 0.05; Y 0.005; OT 0.3; Si 0.1; bal Ti

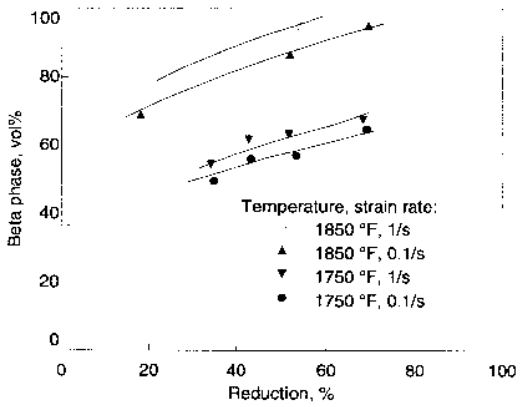
(continued)

Ti-6242: Duplex and triplex anneal strength



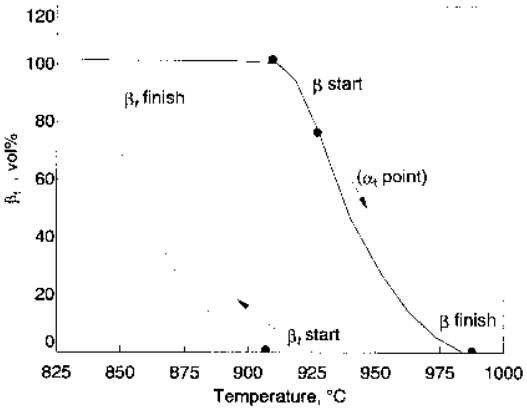
Elongation ductility is similar for both treatments.
Source: R.A. Wood and R.J. Favor, *Titanium Alloys Handbook*, MCIC-HB-02, Battelle Columbus Laboratories, 1972

Ti-6242: Beta fraction vs deformation



Plot of change in volume fraction of beta with deformation.
Source: G.D. Lahoti and T. Altan, "Research to Develop Process Models for Producing a Dual Property Titanium Alloy Compressor Disk," AFWAL-TR-80-4162, 1980

Ti-6242: Transformation hysteresis



Source: G.D. Lahoti and T. Altan, "Research to Develop Process Models for Producing a Dual Property Titanium Alloy Compressor Disk," AFWAL-TR-80-4162, 1980

Heat Treatment

A variety of heat treatments for Ti-6242 are possible. A particular condition is usually selected on the basis of product type, part section size, and properties desired.

Annealing. Several different annealing treatments are available for Ti-6242. Choice depends on the product form and the section size of the product as well as on the properties desired (see tables). A general annealing treatment reported by a producer in the *Titanium Alloys Handbook* (Battelle, 1972) consists of a 1 to 8 h exposure at 705 to 845 °C (1300 to 1500 °F) followed by slow cooling to 565 °C (1050 °F) and subsequent air cooling. For bar and forged sections, solution annealing for 1 h at 900 to 955 °C (1650 and 1750 °F) plus stabilization annealing for 8 h at 595 °C (1100 °F), and air cooling are recommended. In this condition, the guaranteed room-temperature properties of 25 mm (1 in.) diameter bar are as follows:

Ultimate tensile strength	896 MPa (130 ksi)
Tensile yield strength (0.2%)	827 MPa (120 ksi)
Elongation	10%
Reduction of area	25%

Strengthening Heat Treatments. Ti-6242 may be heat treated to obtain higher uniaxial tensile strengths by conventional solution heat treatment and aging exposures. Because the β content of this α-β alloy is small, the response to such a strengthening heat treatment is not great. Furthermore, material in the solution treated and aged condition has lower creep strength than in the annealed-plus-stabilized condition.

Ti-6242: General types of heat treatment

Heat treatment	Temperature °C	Temperature °F	Time, h	Cooling method
Stress relief(a)	480-700	900-1300	0.25-4	AC or slow cool
Annealing		Varies by product form		
Solution treating	955-980	1750-1800	1	WQ
Aging	540-595	1000-1100	8	AC

(a) Stress relief varies depending on the portion of residual stress to be relieved. No data are available, and the given stress relief range covers the values specified in *Metals Handbook*, 9th ed. and *Titanium Alloys Handbook*, Battelle, 1972.

Ti-6242: Annealing treatments for sheet

Treatment	Temperature °C	Temperature °F	Duration, h	Cooling method
Duplex anneal				
1st stage	900	1650	0.5(a)	AC
2nd stage	785	1450	0.25	AC
Triplex anneal(b)				
1st stage	900	1650	2.5(a)	AC
2nd stage	785	1450	0.25	AC
3rd stage	595	1100	2	AC

(a) For sheet less than 3.1 mm (0.125 in.) in thickness, a shorter anneal time at 900 °C (1650 °F) is recommended. Ten minutes has been suggested. (b) The advantage of triplex anneal is higher uniaxial strength.

Ti-6242: Anneal treatments for bar and forgings

Treatment	Temperature °C	Temperature °F	Duration, h	Cooling method
Sections <63.5 mm (2.5 in.) diam				
Anneal	955	1750	1	AC
Stabilization	595	1100	8	AC
Sections >63.5 mm (2.5 in.) diam				
Anneal	900 or 955(a)	1650 or 1750(a)	1	AC
Stabilization	595	1100	8	AC

(a) The 900 °C (1650 °F) treatment, along with the 595 °C (1100 °F) stabilization anneal, provides somewhat higher tensile strengths at room and elevated temperatures, whereas the 955 °C (1750 °F) treatment combined with 595 °C (1100 °F) stabilization as above results in superior creep resistance at the higher temperatures and improved stability. Source: R.A. Wood and R.J. Favor, *Titanium Alloys Handbook*, MCIC-HB-02, Battelle Columbus Laboratories, 1972

that further investigation of disk processing at forging temperatures in the range 900 to 930 °C (1650 to 1705 °F) and heat treatment at temperatures in the range 950 to 970 °C (1740 to 1780 °F) would be worthwhile.

Deformation or heat treatment of the $\alpha + \beta$ and β microstructures above the transus temperature leads to development of transformed β microstructures of Widmanstätten α (slow cooling rate) or martensitic α (fast cooling rate).

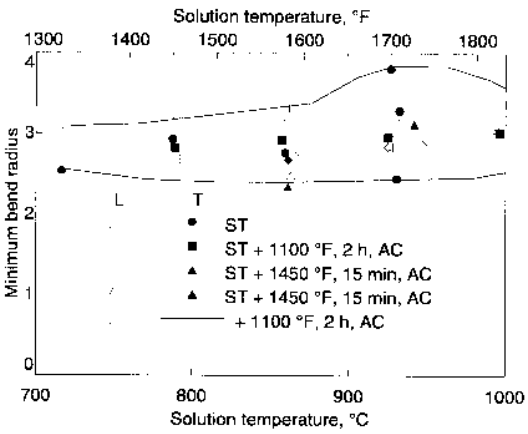
Forming

Ti-6242 Sheet: Minimum bend radii of sheet

Sheet thickness, mm	Sheet thickness, in.	Test direction	Minimum bend radius(a) Duplex anneal	Minimum bend radius(a) Triplex anneal
1.0	0.040	L	3.0	3.0
		T	3.0	3.0
2.0	0.080	L	3.0	3.1
		T	2.8	3.2
3.1	0.125	L	3.3	3.4
		T	3.5	3.4

(a) Minimum bend radius, *r/t*, for 105° after springback. Source: *Aerospace Structural Metals Handbook*, Vol 4, Code 3718, Battelle Columbus Laboratories, 1980

Ti-6242: Minimum bend radii



Effect of solution anneal temperature on room-temperature minimum bend radius of solution annealed, duplex annealed and triplex annealed 0.9 to 1.0 mm (0.036 to 0.040 in.) sheet.
Source: *Aerospace Structural Metals Handbook*, Vol 4, Code 3718, Battelle Columbus Laboratories, 1978

structure leads to grossly nonuniform flow, and at temperatures approaching the β transus, a phenomenon similar to dynamic recrystallization forms primary α. The temperatures at which this latter phenomenon occurs in high-rate deformation are slightly greater than those for low-rate deformation. This trend is analogous to that typically observed in single-phase materials. Heat treatment of high-strain-rate β microstructure compression samples after deformation leads to initial recrystallization in the areas of highest deformation followed by general recrystallization at long times, provided the heat treatment temperature is high enough. The lowest practical temperature for heat treatment of Ti-6242 with a β microstructure is about 915 °C (1680 °F).

The critical strains for die and preform design to make a dual microstructural/dual property disk using β preform material have been suggested by a series of low-strain-rate β microstructure compression tests at 900 °C (1650 °F), followed by heat treatment at 950 °C (1745 °F). The conclusions from these studies are as follows:

- At $\varepsilon < 0.2$, the as-forged microstructures show no distinct transformations. The transformed β structure is retained.
- At $\varepsilon = 0.2$ to 0.4, regions of intense localized shear develop, and these affect the post-deformation heat treatment response.
- At $\varepsilon > 0.4$, regions of dynamic recrystallization, which evolve to the equiaxed α + β structure during heat treatment, are first observed and increase in size with increasing strain.
- Strains $\varepsilon > 1.0$ are required for the equiaxed α + β structure to predominate after heat treatment.

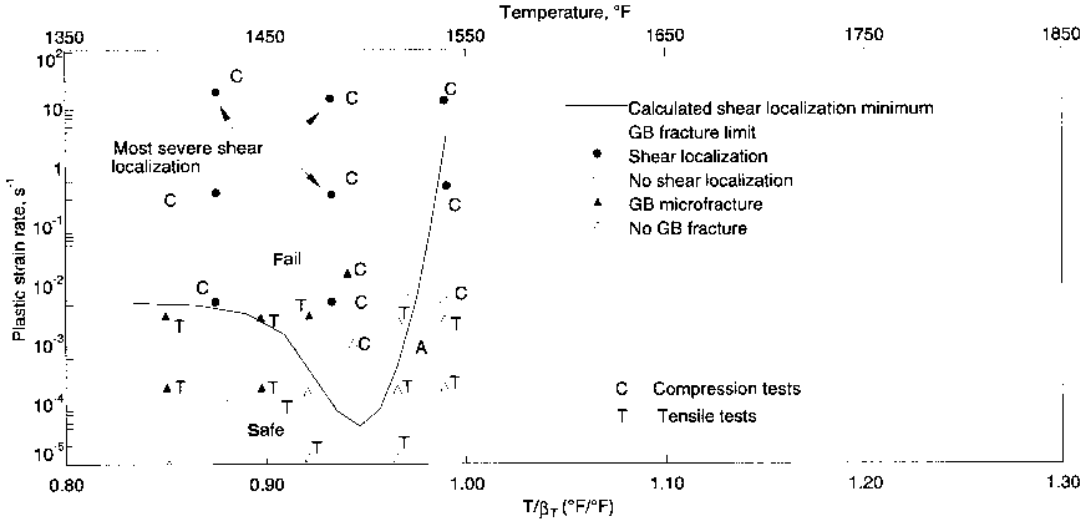
Because complete transformation to the α + β microstructure does not occur even at $\varepsilon = 1.0$, it may be

Ti-6242S: Mechanical properties of alloy forgings

Forge/ heat treat conditions	Creep(a), h to 0.2% strain	Creep rupture time(b), h	K_{Ic} MPa√m	K_{Ic} ksi√in.	RT TYS MPa	RT TYS ksi	RT Elongation, %	480 °C (900 °F) TYS MPa	480 °C (900 °F) TYS ksi	480 °C (900 °F) Elongation, %	Post-creep TYS MPa	Post-creep TYS ksi	Post-creep Elongation, %
(α + β)/β	199	325	81	74	903	131	13	537	78	13	889	129	10
(α + β/β	169	...	~71(c)	~65(c)	924	134	10	565	82	11	...	...	...
(α + β)/(α + β)	56	...	~49(c)	~45(c)	951	138	14	593	86	15	...	...	...
(α + β)/β	19	...	~38(c)	~35(c)	1068	155	3	855	124	5	...	...	...
(α + β)/(α + β)	164	...	~55(c)	~50(c)	896	130	14	565	82	17	...	...	...
β/(α + β)	185	...	~77(c)	~70(c)	930	135	11	579	84	17	...	...	...
β/(α + β)	160	...	~82(c)	~75(c)	937	136	10	...	...	...	...	...	...
β/(α + β)	116	...	83	76	896	130	11	565	82	15	924	134	12
β/β	134	275	79	72	841	122	7	551	80	12	882	128	12

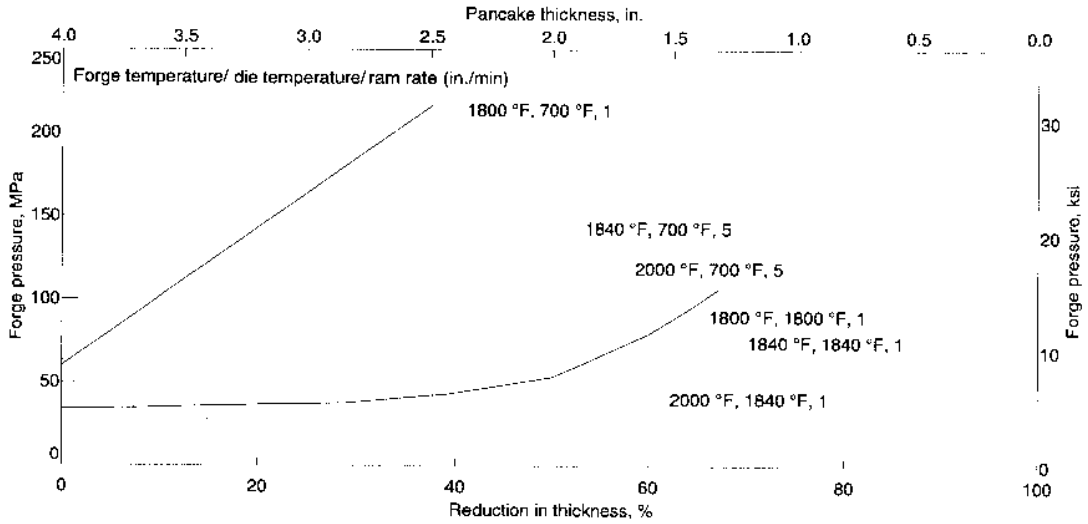
(a) Creep condition: 565 °C (1050 °F), 172 MPa (25 ksi). (b) Creep rupture: 565 °C (1050 °F), 344 MPa (50 ksi). (c) Estimated value. Source: G.D. Lahoti and T. Altan, AFML-TR-79-4156, Dec 1979

Ti-6242: Processing map



The safe region lies below the lines for shear localization and grain boundary (GB) fracture. Shear localization may be stress-state sensitive because it is more likely in compression than tension.
Ti-6242 preform with β microstructure. The region marked A close to the transus is recommended for high-strain-rate deformation.
Source: G.D. Lahoti and T. Altan, AFWAL-TR-81-4130, Oct 1981

Ti-6242: Variation of forge pressure



Variation of forge pressure with reduction for forgings at various forge temperature-die temperature-ram rate combinations.
Source: G.D. Lahoti and T. Altan, AFML-TR-79-4156, Dec 1979

or conventionally forged and supra-transus heat treated product.

Hot die and/or isothermal forging techniques (see Ref 2 for details of these forging processes) are important commercial methods for fabrication of Ti-6242S rotating turbine engine discs to reduce final component cost (from less machining) and/or improve final component microstructural and property uniformity through improved control of forging process conditions. The axisymmetric shapes and designs of such engine components are very well suited to these forging methods. Isothermal forging of Ti-6242S discs is frequently accomplished in a single forging step from the bar or billet sock, under carefully controlled supra- or subtransus metal and die temperatures, levels of strain, and strain rate profiles. Hot die forging, where die temperature approaches but is not equivalent to metal temperature, is also used to reduce unit pressures, enhance forgeability, and produce more sophisticated final shapes in fewer forging operations.

References

1. C.C. Chen, U.S. patent 4,309,226, Jan 1982
2. G.W. Kuhlman, Forging Titanium Alloys, *Metals Handbook*, Vol 10, *Forging and Forming*, ASM International, 1988, p 267-287

Thermomechanical Effects

The effects of heat treatment alone, hot deformation, and the combination of hot deformation and subsequent heat treatment on the microstructures developed in $\alpha + \beta$ and β-preform samples of Ti-6242S were documented in AFWAL-TR-81-4130. The following conclusions were drawn.

Heat treatment of as-received $\alpha + \beta$ microstructure material below the transus temperature leads to minimal changes in the α grain size except at temperatures near the transus (≥954 °C). However, the morphology of the transformed β matrix of this microstructure is a function of heat treatment temperature over the observed range of 900 to 980 °C (1650 to 1800 °F), with higher temperatures leading to a refined matrix phase.

Subtransus heat treatment of as-received β microstructure material appears to lead to negligible changes in microstructure. It is believed that, although there may be α platelet reversion during heat treatment, the basic acicular or lamellar morphology remains the same during heat treatment and thus during cooling back to room temperature after heat treatment.

For the $\alpha + \beta$ microstructure, subtransus deformation or subtransus deformation plus subtransus heat treatment leads to a greater variation of α grain size than observed in the heat treatment studies. It was concluded that structures containing percentages of globular α close to the equilibrium amounts had been developed. This finding is similar to that for previously examined low-strain-rate compression samples and emphasizes the improvement in transformation kinetics brought about by hot work or residual hot work. This hot work also appears to be effective in speeding up the kinetics of the transformation of the β matrix phase when $\alpha + \beta$ compression samples are cooled after hot deformation.

As for low-strain-rate compression, subtransus/high-strain-rate compression of the β micro-

Forging

G.W. Kuhlman, ALCOA, Forging Division

Ti-6242 (and its modification Ti-6242S) is a near-alpha alloy produced commercially in all forging product types, although closed die forgings and rings predominate. The alloy is fabricated using all commercially available types of forging equipment. The silicon-modified version, Ti-6242S, due to its superior creep performance, predominates in commercial forging applications and is a preeminent titanium alloy for elevated-temperature service criteria, particularly rotating and static parts in turbine engines and heated airframe structures. For turbine engine disc forgings, hot die and isothermal forging technologies are important forging fabrication methods, producing near-net closed die forged product with significantly less machining.

Ti-6242S: Forging process temperatures

Process	Metal temperature °C	°F
Conventional forging	900-975	1650-1790
Beta forging	1010-1065	1850-1950

Note: See "Technical Note 4: Forging" for recommended die temperatures.

Forging Process Technology

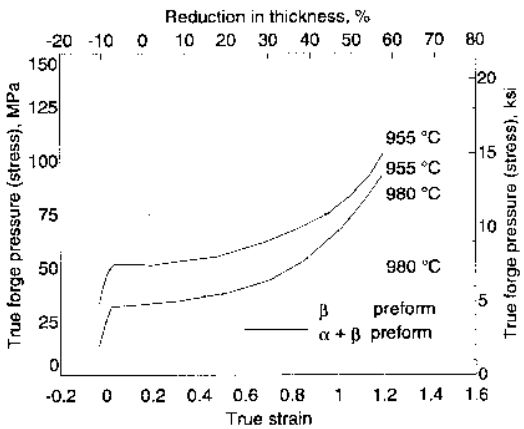
Ti-6242S is characterized by unit pressures (flow stresses), forgeability, and crack sensitivity similar to α–β titanium alloys such as Ti-6Al-4V (see Technical Note 4 and forging of Ti-6Al-4V elsewhere in this volume). The final microstructure of Ti-6242S forgings is developed by thermomechanical processing in forging manufacture tailored to achieve specific microstructural and mechanical property objectives. Thermomechanical processes use combinations of sub- and/or supra-beta transus forging followed by sub- or supra-transus thermal treatments to fulfill critical mechanical property criteria.

Final thermal treatments for forgings include solution treating and aging (stabilizing). Solution treating may be subtransus at 955 °C (1750 °F), followed by air cooling prior to aging, or supra-transus at β_t + 30 to 40 °C (50 to 75 °F), followed by air cooling or faster (e.g., fan cooling, water or oil quenching) cooling rates, prior to aging at 595 °C (1100 °F). Subtransus thermomechanical processes (forging and thermal treatment) for Ti-6242S forgings achieve equiaxed (50 to 75%) α in transformed β matrix microstructures to enhance strength, ductility, and particularly low-cycle fatigue properties. Supra-transus thermomechanical processes (forging and/or thermal treatments) achieve transformed, Widmanstätten and/or colony α microstructures that are exploited for the alloy to enhance creep and fracture-related properties (see Ref 1 in this section).

Conventional Forging. The objectives in forging Ti-6242S are to obtain the final forging shape and desired final microstructure at least cost. Conventional, subtransus (α + β) forging thermomechanical processes currently predominate in commercial forging manufacture. To achieve conventional equiaxed α structures, subtransus reductions of 50 to 75%, accumulated through one or more forging steps, are required. Supra-transus (β) forging for Ti-6242S may be used in early forging operations, including upsetting and open die preforming, to reduce unit pressures and ease forging fabrication. However, higher temperature initial forging operations must be followed by sufficiently subtransus reduction to achieve the desired predominately equiaxed α structure. Conventionally forged Ti-6242S is then subtransus heat treated as noted above.

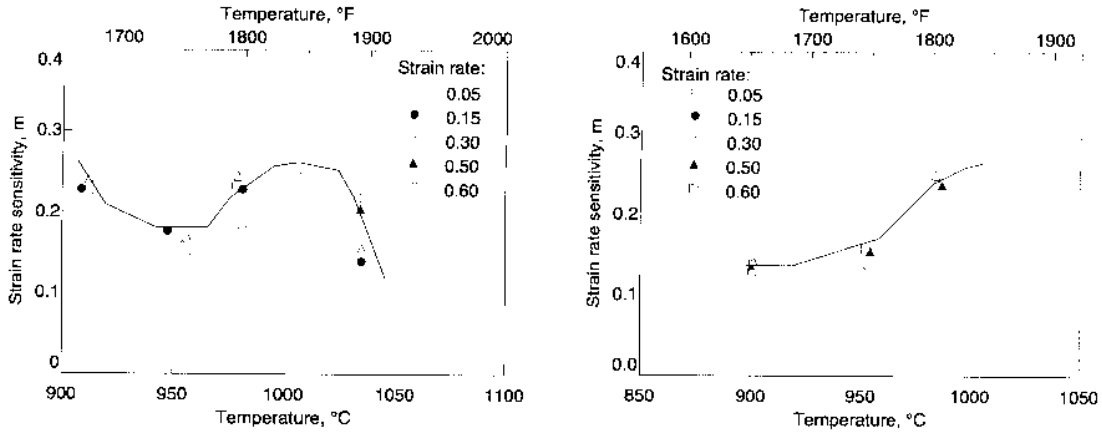
Supra-transus thermomechanical processes for Ti-6242S are becoming commercially significant to achieve transformed Widmanstätten or colony α structures for improved creep and fracture-related properties. Successful β thermomechanical processes for Ti-6242S forgings include both β forging processes followed by subtransus heat treatment or conventional α + β forging followed by supra-transus heat treatment. The latter process provides a more uniform forging product and is currently preferred. Beta forging the alloy requires heavy β reductions (e.g., 35 to 50%) in early forging stages followed by a low final subtransus reduction (e.g., 15 to 25%). Beta forging Ti-6242S requires careful control of forging process conditions, particularly preheat times at temperature, to avoid excessive prior β grain growth. Beta forged Ti-6242S is then subtransus heat treated as noted above. Because of inherent variations in forging conditions, β forged alloy displays more final forging product variation than conventionally subtransus forged and heat treated

Ti-6242: Stress-strain for isothermal forging



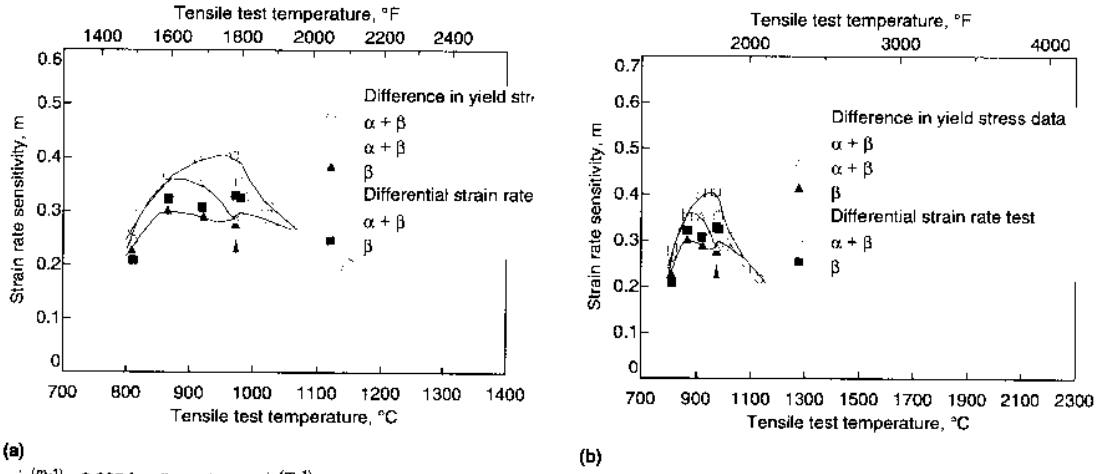
True stress-strain curves for isothermal forging for α + β and β preforms at a strain rate of 5×10^{-3}/s (1 in./min).
Source: C. Chen, RD77-110, Wyman-Gordon, 1977

Ti-6242: Strain-rate sensitivity



Strain-rate sensitivity parameter, m, for (a) $\alpha + \beta$ preform microstructure and (b) β preform microstructure.
Source: S. Semiatin, G.D. Lahoti, and T. Altan, Determination and Analysis of Flow Stress Data for Ti-6242 at Hot Working Temperatures, *Process Modeling Fundamentals and Applications to Metals*, American Society for Metals, 1980, p 387-408

Ti-6242: Strain-rate sensitivity



$\dot{\varepsilon}_1^{(m-1)} = 0.005$ for all specimens; $\dot{\varepsilon}_2^{(m-1)}$ was 0.01 for differential strain rate test and 0.1 to 0.5 for others.
Source: C.C. Chen, "Metallurgical Fundamentals to Ti-6242 (0.1) Alloy Forging," Report RD-77-110, Wyman-Gordon, 1977

dislocation substructure changes at the end of a particular deformation are less than the substructure differences obtained in two separate tests at different nominal strain rates. It is these dislocation substructures and changes in substructure that determine the flow stress and rate sensitivity of the flow stress.

Selected References

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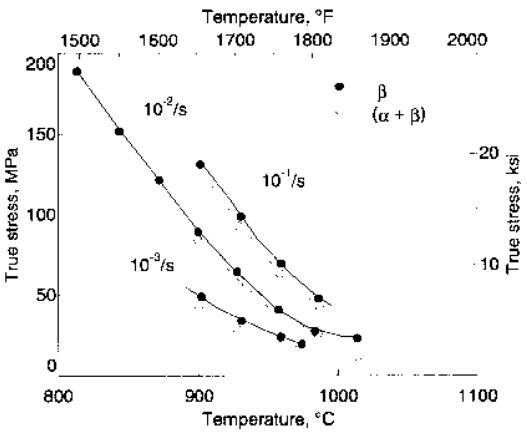
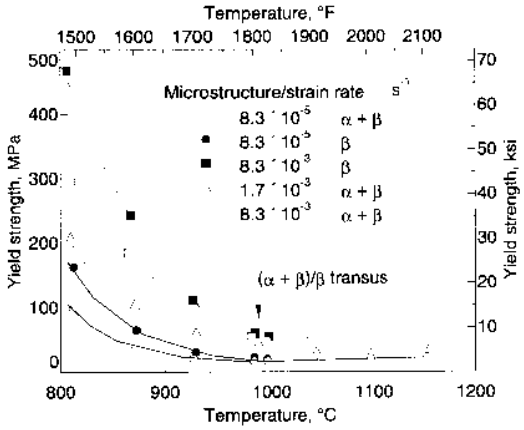
- C.M. Young and O.D. Sherby, Simulation of Extrusion Structures by Means of Torsion Testing for a High Strength Nickel-Base Alloy, Udimet 700, in *Metal Forming—Interrelation Between Theory and Practice*, A.L. Hoffmanner, Ed., Plenum Press, 1971, p 429
- A.J. Griest, A.M. Sabroff, and P.D. Frost, Effects of Strain Rate and Temperature on the Compressive Flow Stresses of Three Titanium Alloys, *Trans. ASM*, Vol 51, 1959, p 935

Ti-6242: Flow stress vs temperature



Temperature dependence of flow stress at different strain rates at ε = 0.6.
Source: G.D. Lahoti and T. Altan, "Research to Develop Process Models for Producing a Dual Property Titanium Alloy Compressor Disk," AFWAL-TR-80-4162, 1980

Ti-6242: Tensile yield strength (0.2%)



Temperature dependence of yield strength for Ti-6242Si alloy.
Source: G.D. Lahoti and T. Altan, AFML-TR-79-4156, Dec 1979

tested at much lower strain rates than those of the present tests. It appears, however, that flow stress is a function of strain-rate history, as evidenced by comparing strain-rate sensitivity data obtained through two different deformation schemes.

Beta Microstructure. The compressive deformation of β specimens gave rise to nonuniform, unstable flow characterized by decreasing load-stroke curves. The stress-strain curves for this microstructure exhibited an additional amount of flow softening in excess of that observed for the α + β microstructure. This increment has been attributed to breakup of the β microstructure and recrystallization to yield the equilibrium α phase at test temperatures below the β transus temperature. It is also believed that development of a softer texture may have caused some of the observed flow softening.

Selected References

- T. Altan and F.W. Boulger, Flow Stress of Metals and Its Application in Metal Forming Analyses, *J. Eng. Ind., Trans. ASME*, Vol 95, 1973, p 1009
- C.C. Chen, "Metallurgical Fundamentals to Ti-6Al-2Sn-4Zr-2Mo-0.1Si Alloy Forgings. I. Influence of Processing Variables on the Deformation Characteristics and the Structural Features of Ti-6242 Si Alloy Forgings," Report RD-77-110, Wyman-Gordon Company, North Grafton, MA, Oct 1977
- C.C. Chen, "Processing, Structure and Properties of Ti-6Al-2Sn-4Zr-2Mo-~0.1Si Alloy Forging," Subcontract Interim Report to Battelle Columbus Laboratories, under Air Force Contract AFML-F-33615-78-C-5025, Aug 1979

Strain-Rate Sensitivity

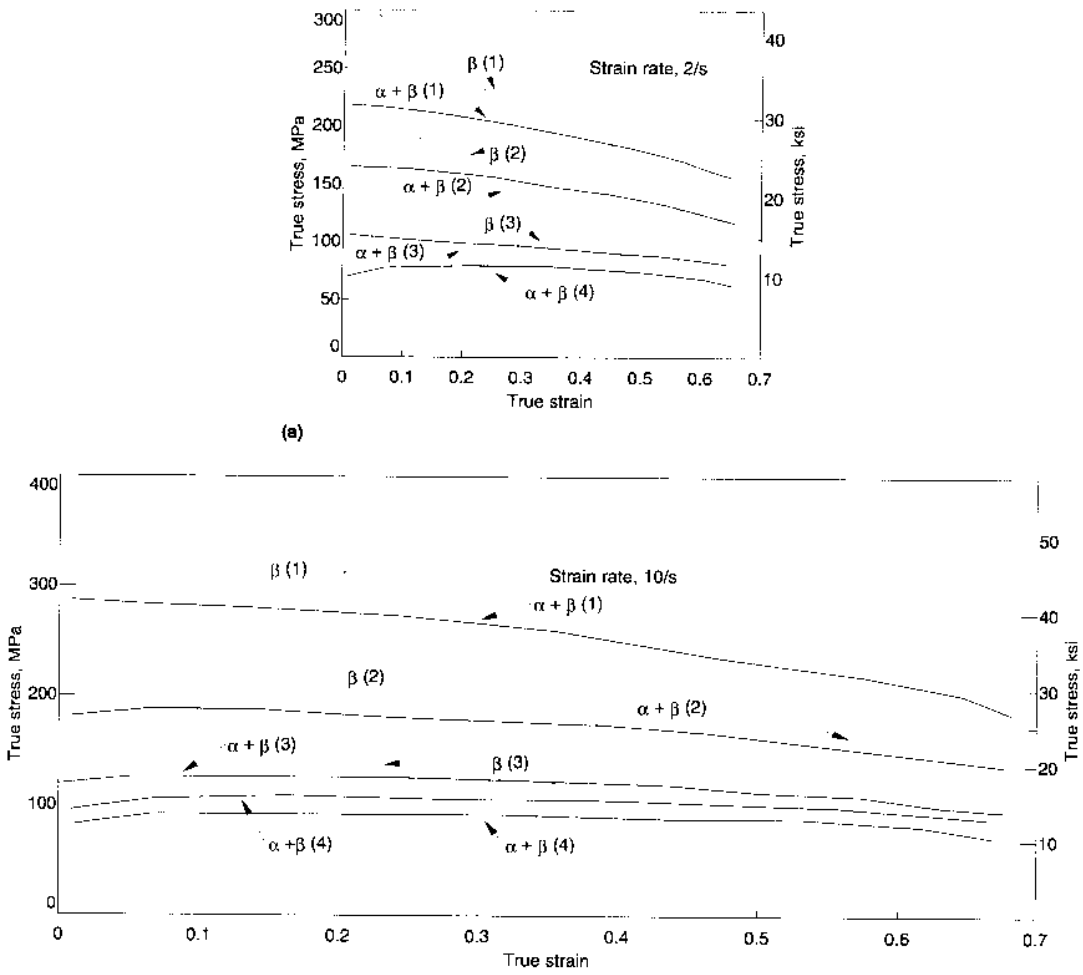
Ti-6242: Strain-rate sensitivity parameter

Microstructure	Temperature °C	Temperature °F	Strain-rate sensitivity (*m*) at: (a)	(b)
α + β	915	1675	0.206	0.181
	955	1750	0.167	0.145
	980	1800	0.201	0.176
	1010	1850	0.259	0.145
	1040	1900	0.217	0.173
β	915	1675	0.134	0.153
	955	1750	0.121	0.156
	980	1800	0.190	0.153

(a) Tests at 2 and 10 s^{-1}; strain = 0.68. (b) Terminal portions of stress-strain data. Average of strain rate = 2 and 10 s^{-1} data. Source: S. Semiatin, G.D. Lahoti, and T. Altan, Determination and Analysis of Flow Stress Data for Ti-6242 at Hot Working Temperatures, *Process Modeling Fundamentals and Applications to Metals*, American Society for Metals, 1980, p 387-408

Determined from tests on nominal strain rates of 2 and 10 s^{-1}, the strain-rate sensitivity parameter, *m*, was found to show a non-negligible temperature dependence (see figures below). The dependence of *m* on strain is weak, except at small strains for the α + β microstructure. The *m* values determined from the terminal portions of stress-strain curves are of similar magnitude as these values. However, they are consistently lower for the α + β microstructure and approximately the same for the β microstructure when compared to measurements from the former method (see table). The lower values for the α + β microstructure can be rationalized from the viewpoint that the

Ti-6242: Softening of α + β and β structures



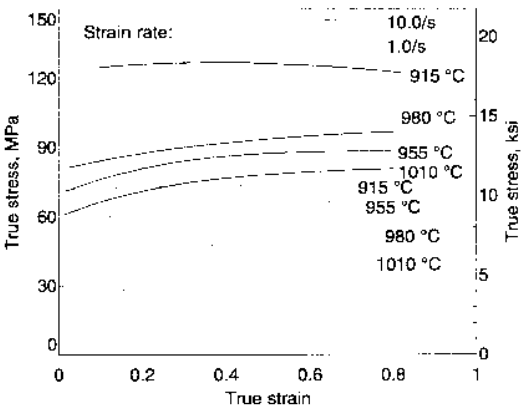
True stress-strain curves at various temperatures. (1) At 915 °C (1675 °F). (2) At 955 °C (1750 °F). (3) At 980 °C (1800 °F). (4) At 1010 °C (1850 °F), 1040 °C (1900 °F). The equilibrium microstructure of equiaxed α in a transformed β matrix deformed stably at all temperatures, but had decreasing stress-strain curves after an initial increase to a peak. In contrast, the nonequilibrium transformed β microstructure deformed unstably at temperatures in the α + β phase field in which it was tested. This instability gave rise to very sharply decreasing stress-strain curves. The flow softening of the equilibrium microstructure has been shown to be largely a result of adiabatic heating. The softening of the nonequilibrium microstructure results from adiabatic heating as well as dynamic recrystallization to produce the (softer) equilibrium microstructure. The development of a softer texture may also contribute to softening of the nonequilibrium microstructure.
Source: S.L. Semiatin *et al.*, in *Process Modeling Fundamentals and Applications to Metals*, American Society for Metals, 1980, p 387-408

Flow Stress

Compression tests run on a mechanical press have been successful in obtaining high-strain-rate compression flow stress data for Ti-6242S (Semiatin *et al.*, *Process Modeling Fundamentals and Applications to Metals*, American Society for Metals, 1980, p 387-480). Data were obtained at hot working temperatures for Ti-6242S having two different microstructures: equiaxed α in a transformed matrix, the α + β microstructure, and a microstructure consisting solely of transformed β, the β microstructure. The flow behavior of the two microstructures differed markedly and are summarized as follows:

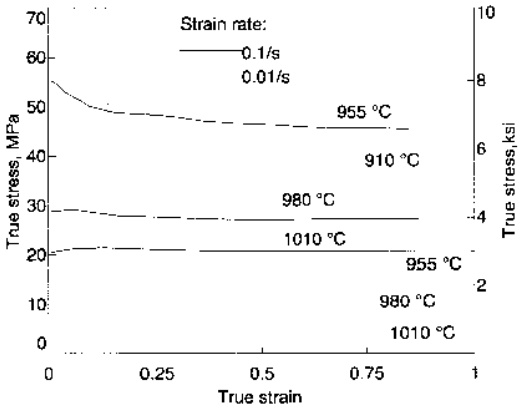
The α + β microstructure deformed stably with monotonically increasing load-stroke curves. The flow behavior exhibited softening (see previous section), which could be attributed to adiabatic heating and a decreasing strain rate as the deformation proceeded, a characteristic of compression tests run in a mechanical press. Adjusting the data for this softening, the dependence of flow stress on strain was found to be weak, and hence, the flow stress is a function primarily of strain rate and temperature. This is a characteristic of metals that dynamically recover during hot working and has been observed in other α + β titanium alloys

Ti-6242: Stress-strain curves up to 1010 °C



Source: G.D. Lahoti and T. Altan, AFML-TR-79-4156, Dec 1979

Ti-6242: Stress-strain curves up to 1010 °C



Source: G.D. Lahoti and T. Altan, "Research to Develop Process Models for Producing a Dual Property Titanium Alloy Compressor Disk," AFWAL-TR-80-4162, 1980

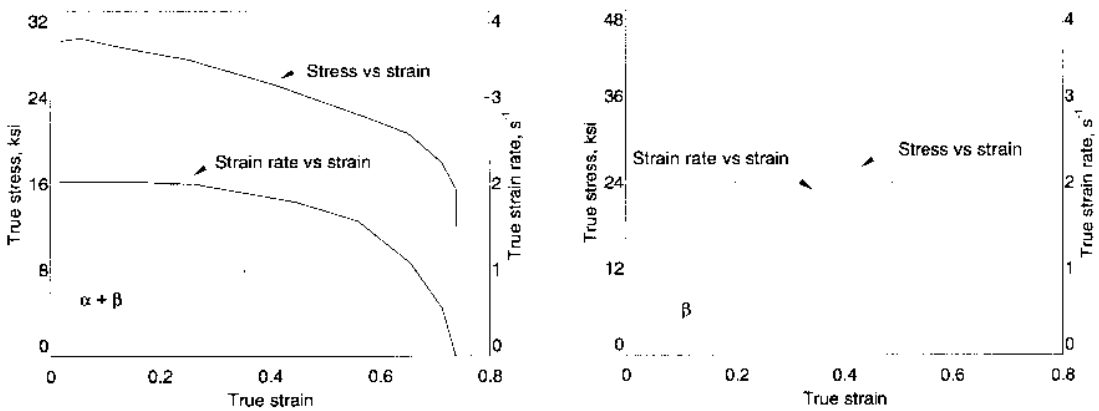
Flow Softening

From flow stress measurements using a mechanical press (Ref 1), measured load-stroke data were reduced to true stress-true strain data assuming frictionless, uniform deformation. This is a good assumption for $\alpha + \beta$ microstructure data, but somewhat approximate for β microstructure data, especially at 915 and 955 °C (1675 and 1750 °F). For both microstructures and strain rates, the stress-strain curves (see figures) decrease monotonically with strain (i.e., flow soften), an effect that is particularly pronounced for β microstructures. It is estimated that the error limits for $\alpha + \beta$ data are +5%, −2% for stresses and ±0.01 for the strains. The flow stress data for this microstructure are similar to those obtained previously in hydraulic and mechanical presses (Ref 2). The error limits for β microstructure are +10%, −2% for stresses, and ±0.01 for strains. However, it is thought that the basic trends in β stress-strain data are correct.

References

1. S.L. Semiatin, G.D. Lahoti, and T. Altan, Determination and Analysis of Flow Stress Data for Ti-6242 at Hot Working Temperatures, *Process Modeling Fundamentals and Applications to Metals*, American Society for Metals, 1980, p 387-408
2. J.R. Douglas and T. Altan, "A Study of Mechanics of Closed-Die Forging (Phase II)," Final Report on Contract No. DAAG46-71-C-0095, Battelle Columbus Laboratories, 1972

Ti-6242: Stress vs strain and strain rate at 915 °C



Source: S.L. Semiatin *et al.*, in *Process Modeling Fundamentals and Applications to Metals*, American Society for Metals, 1980, p 387-408

Ti-6242S: Fracture toughness of forgings from various thermomechanical processing (TMP) routes

Property	TMP route 1(a)	TMP route 2(b)	TMP route 3(c)
Tensile yield strength, MPa (ksi)	937 (136)	903 (131)	917 (133)
Ultimate tensile strength, MPa (ksi)	979 (142)	986 (143)	1006 (146)
Elongation (in 4D), %	16	12	12
Reduction of area, %	34	24	26
Fracture toughness, MPa$\sqrt{m}$ (ksi$\sqrt{in.}$)	56 (51)	78 (71)	76 (69)
Creep at 510 °C, 241 MPa (950 °F, 35 ksi), h to 0.1%	117	447	430
Fatigue crack growth rate ($da/dn \times 10^{-6}$ in./cycle at ΔK 10 ksi$\sqrt{in.}$)	1.2	0.7	0.8

(a) TMP route 1 (baseline): α-β hot die forged plus sub-β transus duplex heat treatment. (b) TMP route 2: β hot die forged plus duplex heat treatment. (c) TMP route 3: β hot die forged plus direct age. Source: G. Kuhlman, T. Yu, A. Chakrabarti, and R. Pishko, "Mechanical Property Tailoring Titanium Alloys for Jet Engine Applications," in *Titanium 1986: Products and Applications*, Titanium Development Association, 1987, p 122

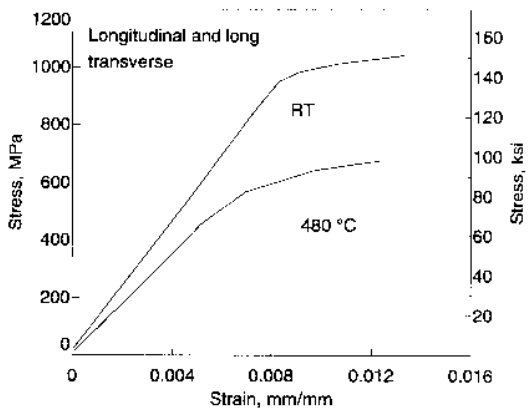
Ti-6242: Fracture toughness of duplex annealed bar

Alloy Condition	Tensile yield strength MPa	Tensile yield strength ksi	Crack plane orientation	Three-point bend specimen: Thickness mm	Thickness in.	Width mm	Width in.	Crack length mm	Crack length in.	K_Q MPa$\sqrt{m}$	K_Q ksi$\sqrt{in.}$
900 °C (1650 °F), 1 h, AC + 595 °C (1100 °F), 8 h, AC	855	124	LT	12.77	0.503	26.47	1.003	13.31	0.524	68(a,b)	62(a,b)
				12.80	0.504	25.42	1.001	12.54	0.494	69(a,b)	63(a,b)
				12.85	0.506	25.50	1.004	13.38	0.527	70(b)	64(b)
	848	123	TL	12.80	0.504	25.50	1.004	12.87	0.507	56(a)	51(a)
				12.75	0.502	25.42	1.001	13.28	0.523	74(b)	68(b)
				12.82	0.505	25.42	1.001	12.98	0.511	61(b)	56(b)
	848	123	TS	12.72	0.501	24.66	0.971	12.42	0.489	60(a)	55(a)
				12.72	0.501	25.65	1.010	12.98	0.511	65(a,b)	59(a,b)
				12.75	0.502	25.65	1.010	13.23	0.521	57(c)	52(c)
975 °C (1790 °F), 1 h, AC + 595 °C (1100 °F), 8 h, AC	820	119	LT	12.80	0.504	25.42	1.001	12.65	0.498	78(a,b)	71(a,b)
				12.80	0.504	25.42	1.001	14.45	0.569	89(b)	81(b)
				12.80	0.504	25.42	1.001	14.91	0.587	70(b)	64(b)
	848	123	TL	12.85	0.506	25.45	1.002	13.53	0.533	82(a,b)	75(a,b)
				12.85	0.506	25.42	1.001	14.09	0.555	77(b)	70(b)
				12.85	0.505	25.42	1.001	13.28	0.523	73(a,b)	67(a,b)
	848	123	TS	12.75	0.502	24.94	0.982	12.72	0.501	67(a,b)	61(a,b)
				12.75	0.502	24.66	0.971	13.66	0.538	71(a,b)	65(a,b)

Note: Specimens were duplex annealed bar 50 by 114 mm (2 by 4.5 in.). Fracture toughness was determined by the three-point bend method. (a) Excessive crack front curvature according to ASTM E-399-74. (b) Insufficient specimen thickness according to ASTM E-399-74. (c) Valid K_{Ic}. Source: C. Hickey, Jr., and T. DeSisto, "Mechanical Properties and Fracture Toughness of Ti-6Al-2Sn-4Zr-2Mo," reported in *Aerospace Structural Metals Handbook*, Vol 4, Code 3718, Battelle Columbus Laboratories, June 1987, p 67

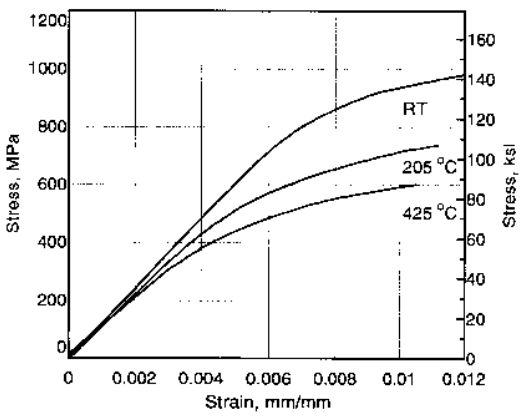
Typical Stress-Strain Curves

Ti-6242: Tensile stress-strain curves



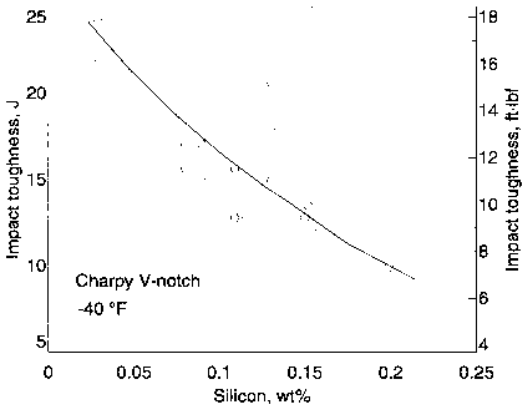
Duplex and triplex annealed 1.2 to 2.1 mm (0.048 to 0.085 in.) sheet after 1/2 h exposure.
Source: MIL-HDBK-5, 1 Dec 1991

Ti-6242: Compressive stress-strain curves



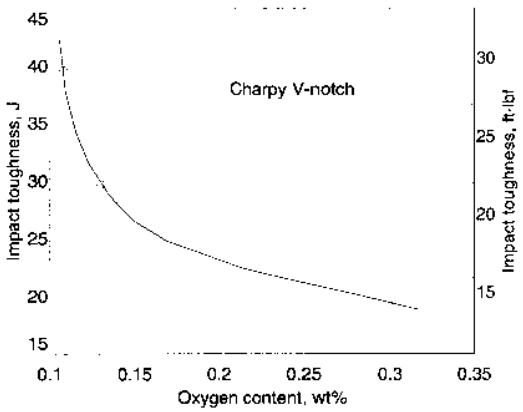
Specimens were cast wedges (tapered plates) and were tested in the as-received, as-cast condition.
Source: O. Deel, "Engineering Data on New Aerospace Structural Materials," AFML-TR-77-198, Battelle Columbus Laboratories, 1977, p 28

Ti-6242: Impact toughness vs silicon content at –40 °C



Effect of silicon content on –40 °C (–40 °F) Charpy-V-notch impact energy of duplex annealed 15.8 mm (5/8 in.) bar treated 15 °C (25 °F) below transus for 1 h, AC, 595 °C (1100 °F), for 8 h, AC.
Source: *Aerospace Structural Metals Handbook*, Vol 4, Code 3718, Battelle Columbus Laboratories, June 1978

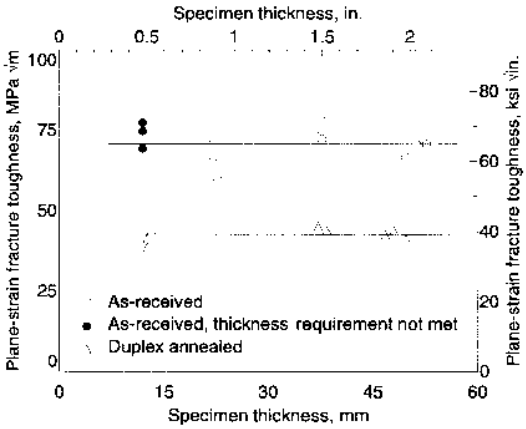
Ti-6242: Effect of oxygen on cast impact toughness



Effect of oxygen content on room-temperature Charpy V-notch impact energy for consumable electrode melted castings. Specimens were consumable electrode skulls melted in water-cooled copper crucibles and cast into tungsten-lined ceramic molds. Standard Charpy V-notch bars cast 0.25 to 0.38 mm (0.010 to 0.015 in.) oversize and machined to final dimensions following heat treatment of 595 °C (1100 °F), 8 h, AC or HeC.
Source: *Aerospace Structural Metals Handbook*, Vol 4, Code 3718, Battelle Columbus Laboratories, June 1978

Fracture Toughness

Ti-6242: Fracture toughness of plate



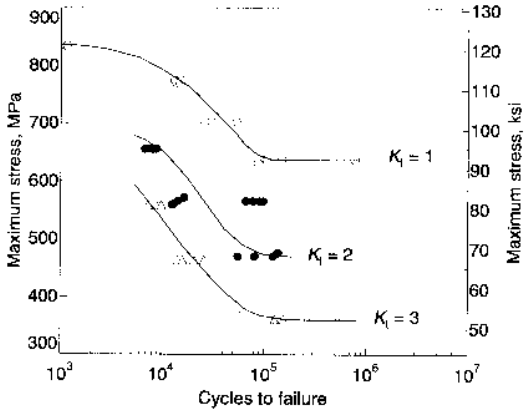
As-received 50 mm (2 in.) plate had a tensile yield strength of 855 MPa (124 ksi). Specimens were duplex annealed at 955 °C (1750 °F) for 30 min, in vacuum, water quenched, then at 455 °C (850 °F) for 17 h, and air cooled. Heat treated specimens had a tensile yield strength of 1041 MPa (151 ksi). Fracture toughness was determined by the three-point bend method, according to ASTM E399-74, except for 13 mm (0.5 in) thick specimens where the thickness requirement was not met. Specimen crack orientation was LT; specimens were symmetrical about the central plane of the plate.
Source: C. Carman, Frankford Arsenal data, 1970; reported in *Aerospace Structural Metals Handbook*, Vol 4, Code 3718, Battelle Columbus Laboratories, 1978, p 67

Ti-6242: Fracture toughness of forgings

Condition	Tensile yield strength, MPa	Tensile yield strength, ksi	Elongation, %	K_{Ic}, MPa√m	K_{Ic}, ksi√in.
α + β forged, βST + aged	903	131	12.5	81	73
β forged, α + βST + aged	896	130	11.0	84	76

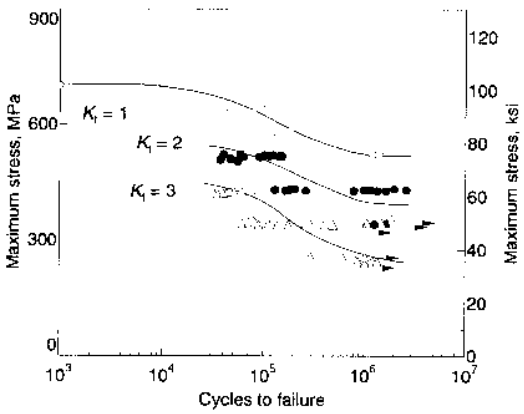
Source: J.C. Williams and E.A. Starke, Jr., The Role of Thermomechanical Processing in Tailoring the Properties of Aluminum and Titanium Alloys, *Deformation, Processing, and Structure*, American Society for Metals, 1984

Ti-6242: Fatigue strength at 200 °C



Specimens were compressor disk forgings (three forgings from three different heats) duplex annealed at 955 °C (1750 °F), 1 h, AC + 595 °C (1100 °F), 8 h, AC. Axial tension, $R = 0$; frequency, 1800 cycles/min.
Source: Pratt & Whitney Data; June 1972; reported in *Aerospace Structural Metals Handbook*, Vol 4, Code 3718, Battelle Columbus Laboratories, 1978, p 84

Ti-6242: Fatigue strength at 455 °C



Compressor disk forgings from three different heats, treated at 955 °C (1750 °F) for 1 h, AC; 595 °C (1100 °F) for 8 h, AC. Axial tension, $R = 0$; frequency, 1800 cycles/min.
Source: Pratt & Whitney Data; June 1972; reported in *Aerospace Structural Metals Handbook*, Vol 4, Code 3718, Battelle Columbus Laboratories, 1978, p 84

Fracture Properties

Impact Toughness

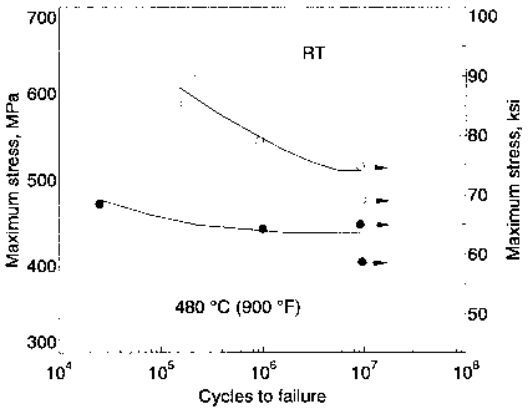
Ti-6242: Effect of heat treatment on RT impact toughness of cast specimens

Heat treatment	Specimen type(a)	Charpy V-notch impact toughness J	ft · lbf
595 °C (1100 °F), 8 h, AC	(b)	30	22
		28	21
		28	21
955 °C (1750 °F), 1 h, AC + 595 °C (1100 °F), 8 h, AC	(b)	38	28
		34	25
		35	26
1035 °C (1900 °F), 1 h, WQ + 595 °C (1100 °F), 8 h, AC	(b)	8	6
		8	6
		8	6
1035 °C (1900 °F), 1 h, HcC + 595 °C (1100 °F), 4 h, HcC	(c) (d)	28	21
	(c) (e)	23-25	17-26

(a) Consumable electrode skull melted in water-cooled copper crucibles and cast into tungsten-lined ceramic molds unless otherwise indicated. (b) Standard Charpy-V bars cast 0.25 to 0.38 mm (0.010-0.015 in.) oversize and machined to final dimensions following heat treatment. (c) Standard Charpy-V bars machined from three thickest sections of heat-treated, variable thickness step castings. (d) Average of nine values from three castings. (e) Range of nine values from three castings. Source: *Aerospace Structural Metals Handbook*, Vol 4, Code 3718, Battelle Columbus Laboratories, June 1978

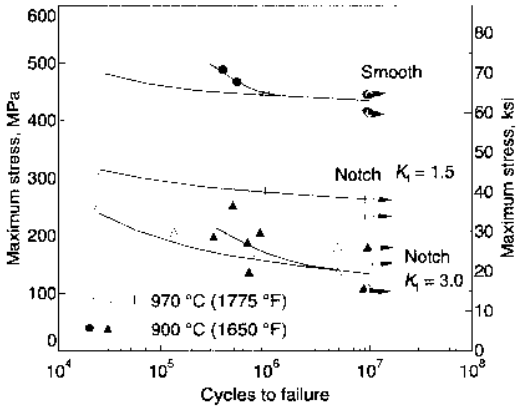
Duplex Annealed Bar

Ti-6242: RT and 480 °C fatigue properties



Specimens were 28.5 mm (1.125 in.) diameter bar duplex annealed at 970 °C (1775 °F), 1 h, AC + 595 °C (1100 °F), 8 h, AC. Ultimate tensile strength, 1006 MPa (146 ksi); tensile yield strength, 958 MPa (139 ksi); smooth, rotating beam tests.
Source: DMIC Data Sheet; Jan 1967, reported in *Aerospace Structural Metals Handbook*, Vol 4, Code 3718, Battelle Columbus Laboratories, 1978, p 83

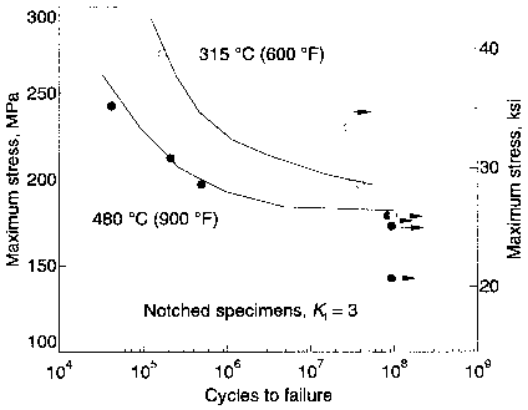
Ti-6242: Fatigue properties at 480 °C



Specimens were 28.5 mm (1.125 in.) bar, duplex annealed as indicated. Rotating beam test results reported at 480 °C (900 °F).
Source: DMIC Data Sheet; Jan 1967; reported in *Aerospace Structural Metals Handbook*, Vol 4, Code 3718, Battelle Columbus Laboratories, 1978, p 84

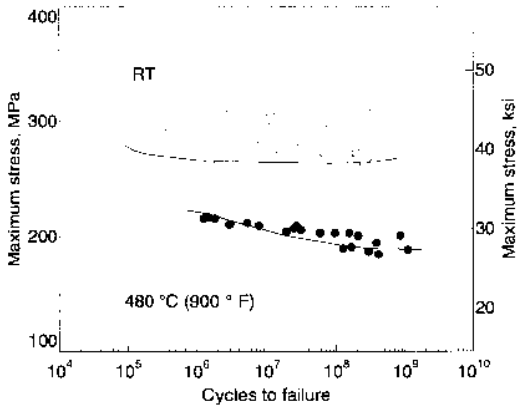
Duplex Annealed Forgings

Ti-6242: Fatigue strength at 315 and 480 °C



Specimens were compressor disk forgings duplex annealed at 955 °C (1750 °F), 1 h, AC + 595 °C (1100 °F), 8 h, AC. Specimens were stress relieved at 540 °C (1000 °F), 2 h in vacuum before testing. Reverse bending (cantilever); $R = -1$; frequency, 120 cycles/s.
Source: Pratt & Whitney Data; June 1972; reported in *Aerospace Structural Metals Handbook*, Vol 4, Code 3718, Battelle Columbus Laboratories, 1978, p 84

Ti-6242: High-frequency fatigue properties

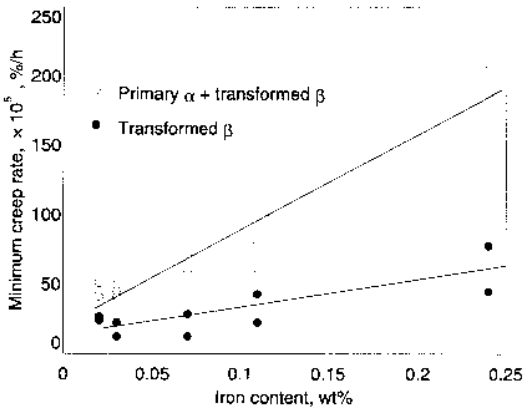


Specimens were 19 mm (0.75 in.) bar duplex annealed at 955 °C (1750 °F), 1 h, AC + 595 (1100 °F), 8 h, AC. Axial fatigue, $R = -1$. Frequency: 13.0 kHz at RT, 13.4 kHz at 480 °C (900 °F). Failure criterion: Crack grown to nearly half of specimen cross section.
Source: NASA TR-72618; July 1969; reported in *Aerospace Structural Metals Handbook*, Vol 4, Code 3718, Battelle Columbus Laboratories, 1978, p 84

Effect of Iron

Iron in the amounts of 0.02 to 0.24 wt% is detrimental to the creep resistance of α–β and β-annealed and stabilized Ti-6242S alloys. The detrimental effects of iron on creep properties are more pronounced in material that has undergone α–β anneal than in β-annealed structures. This is attributed to a change in the predominant creep mechanism, from grain boundary/interface sliding in α-β annealed structures to dislocation creep in β-annealed structures (S. Akem and S.R. Seagle, in *Titanium, Science and Technology,* Vol 4, G. Lütjering, U. Zwicker, and W. Bunk, Ed., Deutsche Gesellschaft für Metallkunde, Germany, 1985, p 2411-2418).

Ti-6242S: Effect of iron on creep rate

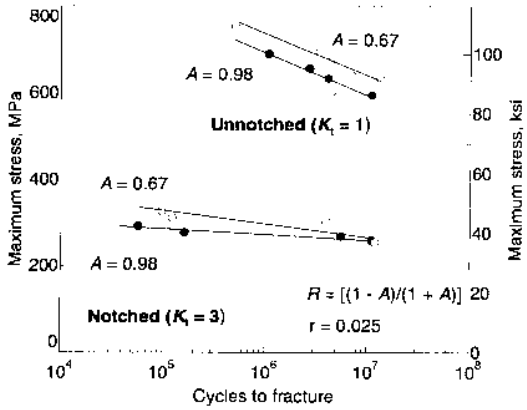


Specimens were tested at 510 °C (950 °F), 241 MPa (35 ksi) for 148 h.
Source: S. Akem and S.R. Seagle, The Detrimental Effect of Iron on Creep of Ti-6242S Alloy, *Titanium, Science and Technology,* Vol 14, G. Lütjering, U. Zwicker, and W. Bunk, Ed., Deutsche Gesselschaft für Metallkunde, Germany, 1985, p 2416

Fatigue Properties

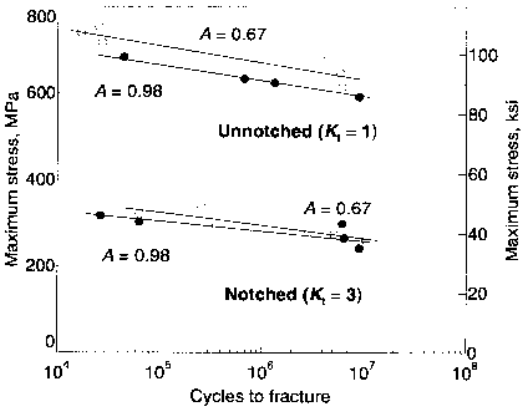
Duplex Annealed Sheet

Ti-6242: RT fatigue properties



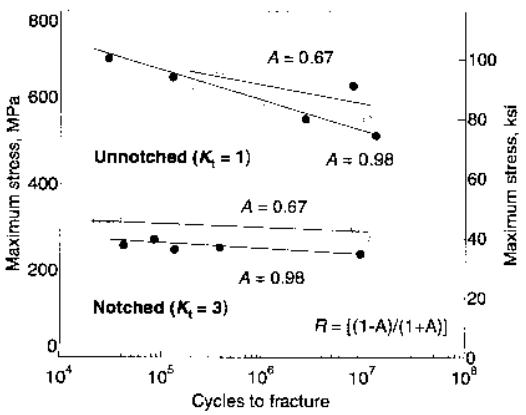
Specimens were 1 mm (0.040 in.) sheet duplex annealed at 900 °C (1650 °F), 30 min, AC + 785 °C (1450 °F), 15 min, AC. Axial fatigue, tension (sinusoidal). Surface: mill finish. Frequency: 2500 cycles/min. Test temperature, 21 °C (70 °F).
Source: AFML-TR-67-41; Apr 1967; reported in *Aerospace Structural Metals Handbook,* Vol 4, Code 3718, Battelle Columbus Laboratories, 1978, p 82

Ti-6242: Fatigue properties at 205 °C



Specimens were 1 mm (0.040 in.) sheet duplex annealed at 900 °C (1650 °F), 30 min, AC + 785 °C (1450 °F), 15 min, AC. Axial fatigue, tension (sinusoidal). Surface: mill finish. Frequency: 2500 cycles/min. Test temperature, 205 °C (400 °F).
Source: AFML-TR-67-41, Apr 1967, reported in *Aerospace Structural Metals Handbook,* Vol 4, Code 3718, Battelle Columbus Laboratories, 1978, p 82

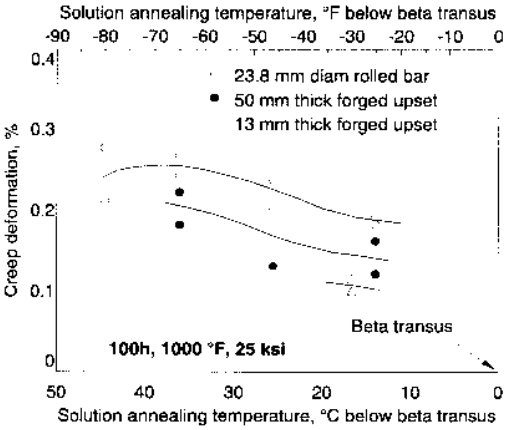
Ti-6242: Fatigue properties at 425 °C



Specimens were 1 mm (0.040 in.) sheet duplex annealed at 900 °C (1650 °F), 30 min, AC + 785 °C (1450 °F), 15 min, AC. Axial fatigue: tension (sinusoidal). Surface: mill finish. Frequency: 2500 cycles/min. Test temperature, 425 °C (800 °F).
Source: AFML-TR-67-41; Apr 1967; reported in *Aerospace Structural Metals Handbook,* Vol 4, Code 3718, Battelle Columbus Laboratories, 1978, p 83

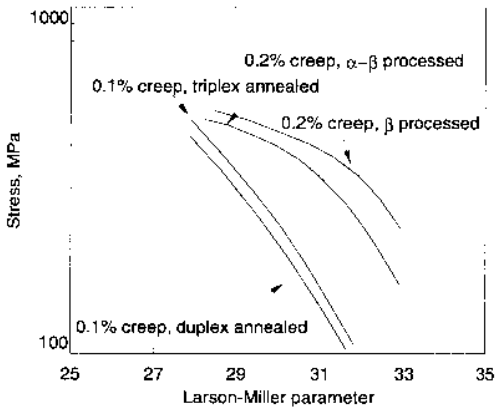
Effect of Processing

Ti-6242S: Effect of solution annealing on creep at 540 °C, 170 MPa



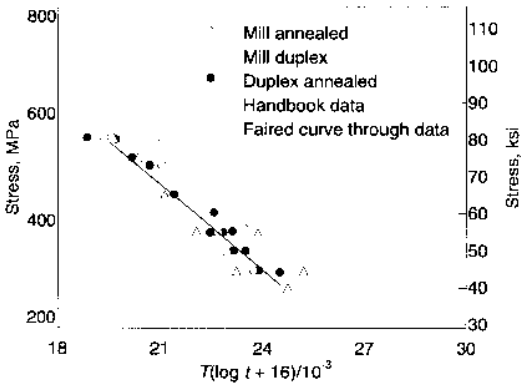
Effect of solution annealing temperature on 100 h, 540 °C (1000 °F), 170 MPa (25 ksi) creep deformation of duplex annealed Ti-6Al-2Sn-4Zr-2Mo-0.09Si rolled bar and forged upsets. Solution treated, 1 h, AC + 595 °C (1100 °F), 8 h, AC.
Source: *Aerospace Structural Metals Handbook*, Vol 4, Code 3718, Battelle Columbus Laboratories, June 1978

Ti-6242: Typical creep curves



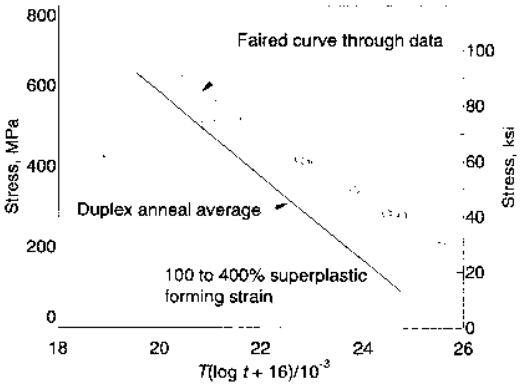
Larson-Miller parameter equals $10^{-3}T(20 + \log t)$, where T is temperature in °R and t is time in hours.
Source: *Metals Handbook*, Vol 3, 9th ed., 1980

Ti-6242: 0.1% creep data for heat treated sheet



Mill annealed sheet of 0.8 mm (0.032 in.) thickness had a chemical composition of 6.0 wt% Al, 0.02 wt% C, 0.06 wt% Fe, 1.9 wt% Mo, 0.010 wt% N, 0.106 wt% O, 0.064 wt% Si, 2.0 wt% Sn, <0.005 wt% Y, and 4.0 wt% Zr. Duplex anneal was carried out at 900 °C (1650 °F) for 15 min, cooled by forced helium gas backfill, then at 785 °C (1450 °F) for 15 min, cooled by the same method. Creep tests were performed according to ASTM recommended practices.
Source: W. Ossa and R. Royster, "Material Characterization of Superplastically Formed Titanium (Ti-6Al-2Sn-4Zr-2Mo) Sheet," NASA Technical Paper 2674, National Aeronautics and Space Administration, 1987, p 26

Ti-6242: 0.1% creep data for superplastically formed sheet



Mill annealed sheet of 0.8 mm (0.032 in.) thickness had a chemical composition of 6.0 wt% Al, 0.02 wt% C, 0.06 wt% Fe, 1.9 wt% Mo, 0.010 wt% N, 0.106 wt% O, 0.064 wt% Si, 2.0 wt% Sn, <0.005 wt% Y, and 4.0 wt% Zr. Duplex anneal was carried out at 900 °C (1650 °F) for 15 min, cooled by the same method.
Superplastically formed panels were produced by axially straining at 785 °C (1650 °F) for various times at a strain rate of 2×10^{-4}/s.
Creep tests were performed according to ASTM recommended practices.
Source: W. Ossa and R. Royster, "Material Characterization of Superplastically Formed Titanium (Ti-6Al-2Sn-4Zr-2Mo) Sheet," NASA Technical Paper 2674, National Aeronautics and Space Administration, 1987, p 26

Creep Rupture

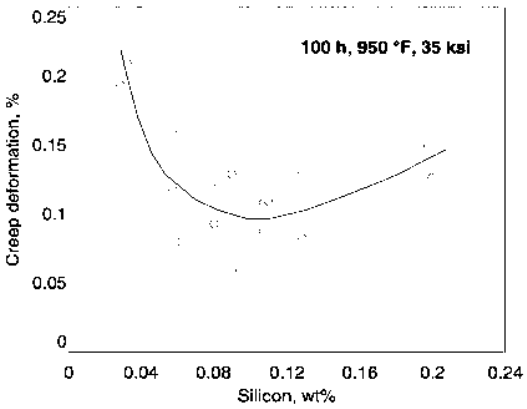
Ti-6242: Creep and rupture properties of castings

Stress MPa	Stress ksi	Temperature °C	Temperature °F	Hours to indicated creep deformation 0.1%	0.2%	0.5%	Initial strain, %	Rupture time, h
689	100	370	700	...	...	...	...	On loading
655	95	370	700	...	...	...	...	On loading
620	90	370	700	270	800	...	2.444	1132.8
551	80	370	700	55	340	4000(a)	1.455	502.7
482	70	370	700	190	4600(a)	...	0.715	1316.2
620	90	480	900	0.02	0.05	0.17	6.430	4.1
551	80	480	900	0.9	3.0	20	0.963	544.0
413	60	480	900	4.5	22	232	0.452	626.2
310	45	480	900	14	78	1030(a)	0.474	649.1
241	35	480	900	165	975	4300(a)	0.018	955.7

Note: Chemical composition was 6.02 wt% Al, 0.018 wt% C, 0.010 wt% Fe, 0.0047 wt% H, 2.07 wt% Mo, 0.013 wt% N, 0.168 wt% O, 0.05 wt% Si, 2.04 wt% Sn, and 3.80 wt% Zr. Alloy was tested in as-received condition, 150 by 165 mm (6 by 6.5 in.) cast wedges (tapered plate, 25 to 6.4 mm, or 1 to 0.25 in.). Source: O. Deel, "Engineering Data on New Aerospace Structural Materials," AFML-TR-77-198, Battelle Columbus Laboratories, 1977, p 25, 33. (a) Estimated

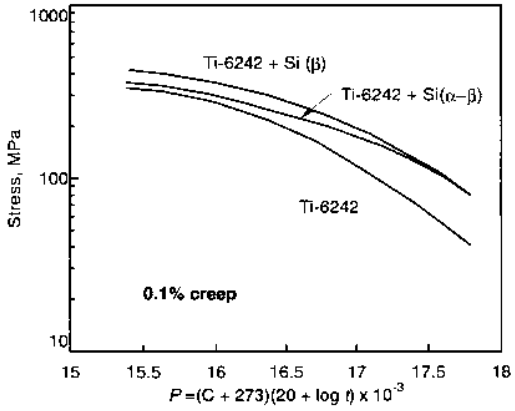
Effect of Silicon

Ti-6242: Effect of silicon on creep at 510 °C, 240 MPa



Specimens were standard composition except for silicon. 15.8 mm (0.625 in.) bar treated 14 °C (25 °F) below the transus, 1 h, AC + 595 °C(1100 °F), 8 h, AC.
Source: *Aerospace Structural Metals Handbook*, Vol 4, Code 3718, Battelle Columbus Laboratories, June 1978

Ti-6242: Larson-Miller plots of two Si levels

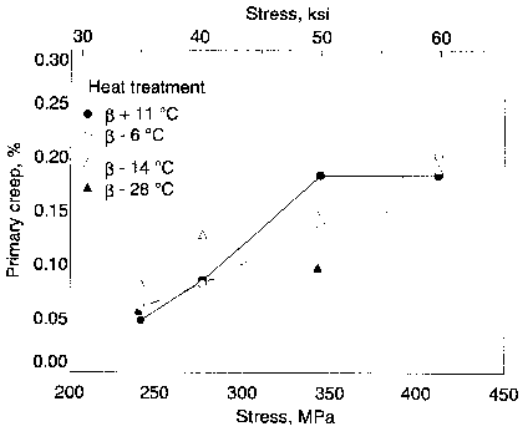


Condition for Ti-6242 without silicon is unspecified
Source: E. Huron and J. Miller, "Titanium Requirements for Current and Future Military Gas Turbine Engines," in *Space Age Metals Technology*, F. Froes and R. Cull, Ed., SAMPE, 1989, p 271

Effect of Nickel

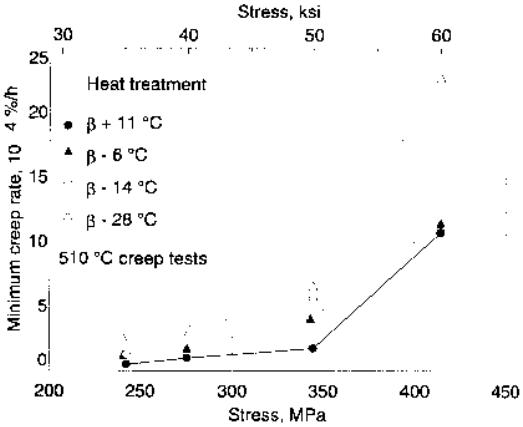
Small concentrations of nickel appear to segregate uniformly into the beta phase and dramatically degrade the creep properties of Ti-6242S (*Met Trans*, Vol 24A, Aug 1993, p 1819-1826). The explanation for this is unclear.

Ti-6242S: Effect of heat treatment on primary creep at 510 °C



Primary creep vs applied stress for 510 °C (950 °F) creep tests of specimens solution treated at four temperatures, followed by air cooling and a stabilization age of 600 °C (1110 °F) for 8 h.
Source: P. Bania and J. Hall, Creep Studies of Ti-6242-Si Alloy, in *Titanium, Science and Technology*, Vol 4, G. Lütjering, U. Zwicker, and W. Bunk, Ed., Deutsche Gesellschaft für Metallkunde, Germany, 1985, p 2371-2378

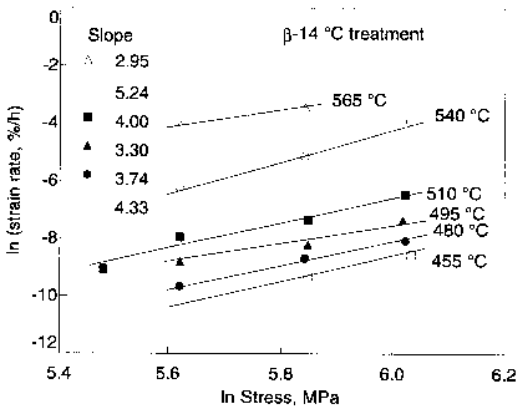
Ti-6242S: Effect of heat treatment on steady state creep rates



Steady-state (minimum) creep rate vs. applied stress for 510 °C (950 °F) creep tests of specimens solution treated at four temperatures, followed by air cooling and a stabilization age of 600 °C (1100 °F) for 8 h.
Source: P. Bania and J. Hall, Creep Studies of Ti-6242-Si Alloy, *Titanium, Science and Technology*, Vol 4, G. Lütjering, U. Zwicker, and W. Bunk, Ed., Deutsche Gesellschaft für Metallkunde, Germany, 1985, p 2371-2378

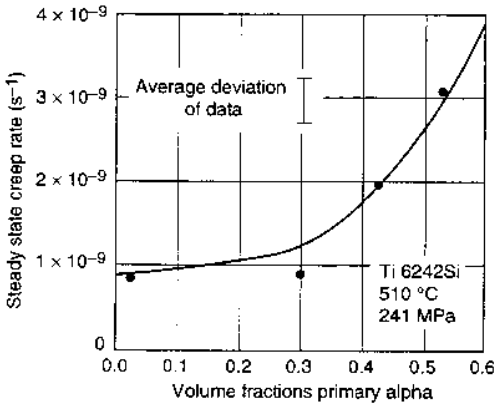
Creep Rates

Ti-6242S: Minimum creep rates at various temperatures



Plot of ln $\dot{\varepsilon}$ and ln (applied stress) for solution treated and stabilized Ti-6242 Si. Solution treatment was 14 °C (25 °F) below the β transus and for 1 h, followed by air cooling and aging at 600 °C (1110 °F) for 8 h.
Source: P. Bania and J. Hall, Creep Studies of Ti-6242-Si Alloy, *Titanium, Science and Technology*, Vol 4, G. Lütjering, U. Zwicker, and W. Bunk, Ed., Deutsche Gesellschaft für Metallkunde, Germany, 1985, p 2371-2378

Ti-6242S: Effect of primary alpha on creep rate



Source: *Met. Trans*, Vol 24A, August 1993, p 1819-1826

Ti-6242: Creep and creep rate data of castings

Stress MPa	Stress ksi	Temperature °C	Temperature °F	Hours to indicated creep deformation: 0.1%	0.2%	0.5%	1.0%	2.0%	Initial strain, %	Minimum creep rate, %/h
689	100	370	700	...	...	...	...	...	...	...
655	95	370	700	...	...	...	...	...	...	...
620	90	370	700	270	800	...	...	...	2.444	0.0002
551	80	370	700	55	340	4000(a)	...	...	1.455	0.00008
482	70	370	700	190	4600(a)	...	...	...	0.715	0.00001
620	90	480	900	0.02	0.05	0.17	0.47	1.5	6.430	0.92
551	80	480	900	0.9	3.0	20	66	154	0.963	0.010
413	60	480	900	4.5	22	232	680	1600(a)	0.452	0.0011
310	45	480	900	14	78	1030(a)	...	...	0.474	0.00024
241	35	480	900	165	975	4300(a)	...	...	0.018	0.00009

As-received cast wedges. (a) Estimated. Source: AFML-TR-77-198

Creep Properties

Studies on the creep behavior of Ti-6242-Si by Bania and Hall (see table) indicate the presence of two creep mechanisms, although the actual mechanisms are not clearly identified. This reinforces the argument that creep testing for design data generation or qualification testing must be performed under conditions (temperature and stress) similar to those anticipated in service. Otherwise, metallurgical changes that affect one mechanism and not another could be overlooked. Other conclusions of the study by Bania and Hall are as follows. Primary creep in Ti-6242-Si behaves in an irregular manner and is not easily correlated with microstructure. The steady-state creep rate correlates well with microstructure, indicating a direct proportionality with primary α content. Work by Thiehson *et al.* (*Met. Trans.* Vol 24A, Aug 1993, p 1819-1826) did show a correlation of both the steady-state and primary creep rates with microstructure. They also reported that Ni up to 0.1 wt.% had a deleterious effect on the creep rate, while Cr contents up about 0.3 wt.% had very little, if any, effect.

Much of the scatter in qualification testing can be attributed to varying contributions of erratic primary creep and microstructural variations. The fully transformed Ti-6242-Si β structure is superior to the conventional (19% equiaxed primary α) structure by roughly 20 °C (36 °F) or 70 MPa (10 ksi). Two creep mechanisms appear to be active: a high-activation energy (84 to 100 kcal/mol) process in the 495 to 565 °C (920 to 1045 °F) range and a low-activation energy (12 to 25 kcal/mol) process in the 455 to 480 °C (850 to 895 °F) range. Qualification creep testing should be conducted at temperatures typical of the intended use temperature and at stresses that minimize the impact of primary creep.

Ti-6242S: Creep test data

Test temperature °C	Test temperature °F	Stress MPa	Stress ksi	Primary creep, %	Steady-state creep, 10^4 %/h
0% primary α					
455	850	345	50	0.050	0.52
480	895	345	50	0.070	1.10
510	950	241	35	0.045	0.73
		276	40	0.083	1.07
		345	50	0.185	1.48
		414	60	0.182	10.7
540	1000	345	50	0.110	18.4
565	1045	345	50	0.10	137.0
7% primary α (average)					
510	950	241	35	0.06	1.4
		276	40	0.08	1.5
		345	50	0.135	4.2
		414	60	0.20	11.5
35% primary α (average)					
510	950	241	35	0.055	2.4
		276	40	0.085	3.0
		345	50	0.095	6.6
		414	60	0.190	23.0
19% primary α (average)					
455	850	276	40	0.06	0.36
		345	50	0.055	0.84
		414	60	0.095	2.05
480	895	276	40	0.045	0.65
		345	50	0.08	1.60
		414	60	0.11	2.9
495	920	276	40	0.08	1.7
		345	50	0.115	2.65
		414	60	0.14	6.50
510(a)	950(a)	241	35	0.075	1.30
		276	40	0.091	3.30
		345	50	0.116	5.85
		414	60	0.122	13.1
540	1000	276	40	0.11	21.2
		345	50	0.10	55.0
		414	60	0.20	174.0
565	1045	276	40	0.10	184.0
		345	50	0.20	349.0

(a) Average of eight tests; all other values represent an average of two tests. Source: P. Bania and J. Hall, Creep Studies of Ti-6242-Si Alloy, in *Titanium, Science and Technology*, Vol 4, Proc. of Fifth International Conference on Titanium, G. Lütjering, U. Zwicker, and W. Bunk, Ed., Deutsche Gesellschaft für Metallkunde, Germany, 1985, p 2371-2378

Ti-6242: Observed stability of ductility in rolled bar after 1000-h exposure

10000-h exposure at					Subsequent RT tensile properties:					
Temperature		Stress		Defor-	Ultimate tensile strength		Tensile yield strength		Elongation in 25 mm (1 in.),	Reduction of
°C	°F	MPa	ksi	mation, %	MPa	ksi	MPa	ksi	%	area, %
900 °C (1650 °F), 1 h, AC + 595 °C (1100 °F), 8 h, AC										
		None			944	137.0	832	120.7	15.5	31.2
425	800	379	55	0.11	1016	147.4	912	132.3	15.5	34.3
455	850	344	50	0.15	990	143.6	912	132.3	18.5	34.8
480	900	275	40	0.17	982	142.4	890	129.1	17.5	28.9
540	1000	103	15	0.12	979	142.1	904	131.2	15.0	31.6
955 °C (1750 °F), 1 h, AC + 595 °C (1100 °F), 24 h, AC										
		None			962	139.5	857	124.3	17.5	32.8
425	800	448	65	0.22	942	136.7	849	123.2	18.0	28.2
455	850	344	50	0.14	959	139.2	862	125.0	20.5	32.2
		379	55	0.18	941	136.5	846	122.7	18.0	30.2
480	900	275	40	0.22	942	136.6	850	123.3	19.0	30.2
		310	45	0.23	952	138.1	865	125.5	19.5	33.5
540	1000	138	20	0.17	957	138.9	888	128.9	16.5	30.8
1025 °C (1875 °F), 1 h, AC + 595 °C (1100 °F), 24 h, AC										
		None			954	138.4	834	121.0	15.0	21.2
425	800	448	65	0.19	955	138.5	869	126.1	13.5	23.4
455	850	344	55	0.14	962	139.6	867	125.8	14.5	21.9
480	900	310	45	0.16	964	139.8	873	126.7	14.5	23.4
540	1000	138	20	0.10	977	141.7	894	129.7	12.5	18.3

Note: Specimens were 57 mm (2.25 in.) diam rolled bar duplex annealed in full sections as indicated. Subsequent tensile properties were measured in the outside longitudinal direction and were tested after exposure without surface conditioning. Source: *Aerospace Structural Metals Handbook*, Vol 4, Code 3718, Battelle Columbus Laboratories, June 1978

Ti-6242: Loss of ductility after exposure at 425 °C

Effect of long-time exposure to elevated temperature with load on room-temperature tensile properties of duplex annealed pancake forgings

Exposure conditions							Subsequent RT tensile properties:					
	Temperature		Stress		Time,	Plastic deformation,	Ultimate tensile strength		Tensile yield strength		Elongation,	Reduction of area,
Heat	°C	°F	MPa	ksi	h	%	MPa	ksi	MPa	ksi	%	%
A			Unexposed				993-1034(a)	144-150(a)	903-965(a)	131-140(a)	14-17(a)	42-51(a)
B			Unexposed				958-999(b)	139-145(b)	868-917(b)	126-133(b)	11-17(b)	33-52(b)
A	370	700	689	100	1500	1.681	1068	155	1061	154	14	44
B			551	80	1500	0.221	1013	147	979	142	18	39
B			413	60	1500	0.221	993	144	896	130	17	39
A	425	800	620	90	1500	1.444	1061	154	1048	152	14	28
A			586	85	1000	0.713	1061	154	1034	150	8	10
A			551	80	1000	0.433	1034	150	1013	147	15	44
A			517	75	500	0.256	1034	150	979	142	8	15
B			344	50	1500	9.158	986	143	910	132	16	39
B	480	900	344	50	1500	0.592	965	140	917	133	17	39
B			207	30	1500	0.173	1006	146	965	140	9	19
B	540	1000	207	30	1500	7.706	930	135	882	128	9	19
B			138	20	1500	2.051	1013	147	924	134	14	21

Note: Specimens were duplex annealed at 955 °C (1750 °F), 1 h, AC + 595 °C (1100 °F), 8 h, AC. (a) Range of values from six tests on two forgings. (b) Range of values from twelve tests on six forgings. Source: *Aerospace Structural Metals Handbook*, Vol 4, Code 3718, Battelle Columbus Laboratories, June 1978

(900 °F), 137 MPa (20 ksi) for duplex annealed sheet and at 540 °C (1000 °F), 69 MPa (10 ksi) for triplex annealed sheet. At 510 °C (950 °F), 103 MPa (15 ksi), no damage is done to triplex annealed sheet for 1000-h exposure. It appears that the triplex annealed condition is more stable than the duplex condition.

No change in smooth tensile properties at the exposure temperature is observed after 1000-h exposure without load up to 540 °C (1000 °F) and perhaps beyond. However, the influence of exposure to elevated temperature without load on the subsequent room-temperature sharp notch strength of sheet is a sensitive indicator of instability and reveals the onset of serious embrittlement at 480 °C (900 °F) for 1000-h exposure for both duplex and triplex annealed conditions (see the previous section "Notched Tensile Strength").

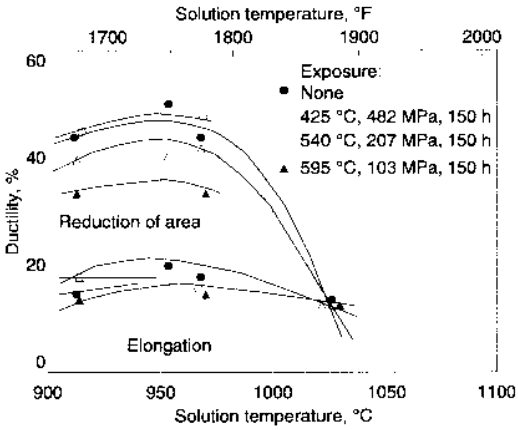
Bar in the recommended duplex annealed condition exposed 1000 h with load at temperatures up to 540 °C (1000 °F) exhibits no significant instability (see table). However, losses in ductility are observed for forgings in the recommended duplex annealed condition at exposure temperatures as low as 425 °C (800 °F).

perature smooth tensile properties is observed, whereas at 206 MPa (30 ksi) a slight loss in ductility is observed (see figure). Bar rolled in the β region is unaffected by 150-h exposure at 540 °C (1000 °F), 206 MPa (30 ksi).

Forged bar in the recommended duplex annealed condition exposed 150 h up to 540 °C (1000 °F), 172 MPa (25 ksi) exhibits no loss in subsequent room-temperature smooth tensile properties. Contrasting results were obtained from forged compressor wheels. The hub location in one wheel exposed at 425 °C (800 °F), 482 MPa (70 ksi), 150 h showed losses in elongation and reduction of area. Other locations in the same forging, however, were essentially unaffected. A second wheel forging, water quenched from the solution anneal, exhibited losses in ductility after exposures at 480 °C (900 °F)/379 MPa (55 ksi), 540 °C (1000 °F)/241 MPa (35 ksi), or 595 °C (1100 °F)/103 MPa (15 ksi).

Long-Time Stability. It has been reported that 1000-h exposure up to 540 °C (1000 °F) without load has no deleterious effect on subsequent room-temperature tensile properties of duplex annealed sheet. In contrast, 1000-h exposure, with load produces a serious loss in ductility at 480 °C

Ti-6242: Ductility loss after high-temperature exposure



13 mm (0.5 in.) diameter round bar, α + β rolled and solution treated (ST) at indicated temperatures for indicated times + 595 °C (1100 °F), 8 h, AC.
Source: *Aerospace Structural Metals Handbook*, Vol 4, Code 3718, Battelle Columbus Laboratories, June 1978

Ti-6242: Effect of 150-h exposure on ductility of duplex annealed compressor hub forging

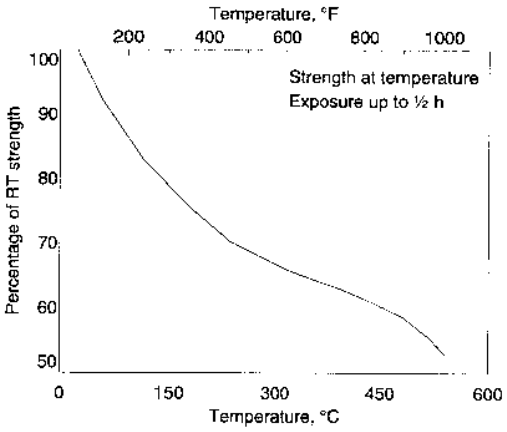
Location	150-h exposure at: Temperature °C	°F	Stress MPa	ksi	Creep in 150 h, %	Ultimate tensile strength MPa	ksi	Tensile yield strength MPa	ksi	Elongation, %	Reduction of area, %
Rim	None		None		...	965	140.0	893	129.6	13.5	28.6
Web	None		None		...	987	143.2	899	130.4	13.5	32.7
Hub	None		None		...	966	140.2	867	125.8	18.0	42.3
Rim	425	800	482	70	0.149	973	141.1	915	132.8	16.0	30.1
Web	425	800	482	70	0.160	963	139.7	902	130.9	16.0	28.1
Hub	425	800	482	70	0.148	955	138.5	899	130.5	15.0	28.8
Rim	540	1000	241	35	0.412	994	144.2	894	129.7	12.5	32.8
Web	540	1000	241	35	0.424	1001	145.2	921	133.6	17.0	29.4
Hub	540	1000	241	35	0.498	955	138.5	871	126.4	12.5	34.1

Note: Specimens were 470 mm (18.5 in.) diam compressor hub forgings duplex annealed at 955 °C (1750 °F), 1 h, AC + 595 °C (1100 °F), 8 h, AC tested in the radial direction. Source: *Aerospace Structural Metals Handbook*, Vol 4, Code 3718, Battelle Columbus Laboratories, June 1978

Ti-6242: Ductility loss after 150-h exposure of duplex annealed compressor wheel forging

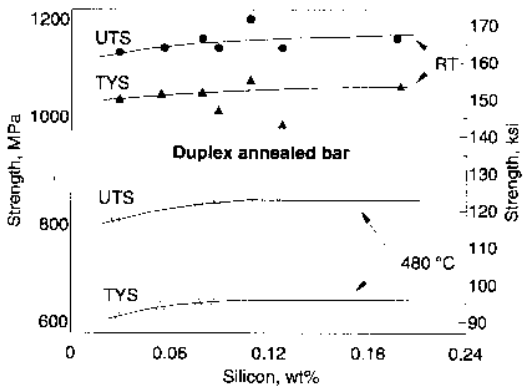
150-h exposure at: Stress MPa	ksi	Temperature °C	°F	Plastic deformation(a), %	Subsequent RT tensile properties: Ultimate tensile strength MPa	ksi	Tensile yield strength MPa	ksi	Elongation, %	Reduction of area, %
Duplex annealed: 900 °C (1650 °F), 1 h, AC + 595 °C (1100 °F), 8 h, AC										
		None			1055	153.0	964	139.8	16.0	37.3
		None			1016	147.4	937	136.0	17.0	44.2
448	65	455	850	0.15	975	141.5	889	129.0	19.0	42.0
379	55	480	900	0.19	973	141.2	888	128.8	18.0	43.6
310	45	510	950	0.24	981	142.3	893	129.6	20.0	42.0
241	35	540	1000	0.30	983	142.7	891	129.3	19.0	47.6
103	15	595	1100	0.54	979	142.0	911	132.2	17.0	34.1
STA 955 °C (1750 °F), 1 h, WQ + 595 °C (1100 °F), 8 h, AC										
		None			1130	164.0	1016	147.4	14.0	41.9
		None			1150	166.9	1042	151.2	14.0	37.8
448	65	455	850	0.26	1136	164.8	1070	155.2	13.0	40.8
379	55	480	900	0.39	1139	165.2	1030	149.4	13.0	34.0
310	45	510	950	0.49	1135	164.6	1008	146.3	14.0	38.3
241	35	540	1000	0.81	1133	164.4	1032	149.7	11.0	31.5
103	15	595	1100	1.04	1110	161.0	1024	148.6	11.0	23.2

Ti-6242: Compressive strength vs temperature



Duplex annealed sheet.
Source: MIL-HDBK-5, 1 Dec 1991

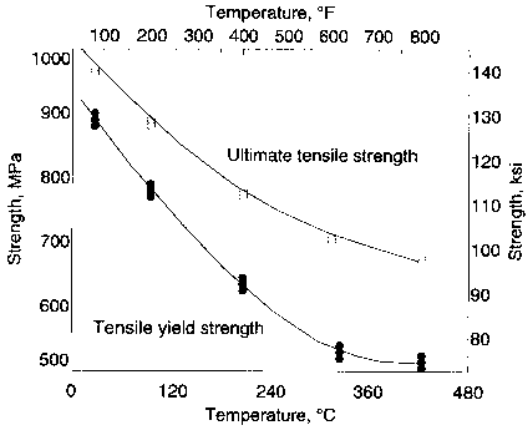
Ti-6242: Tensile properties of bar vs silicon content



Specimens were standard composition except for silicon. 15.8 mm (0.625 in.) bar heat treated 14 °C (25 °F) below the transus, 1 h, AC + 595 °C (1100 °F), 8 h, AC.
Source: *Aerospace Structural Metals Handbook*, Vol 4, Code 3718, Battelle Columbus Laboratories, June 1978

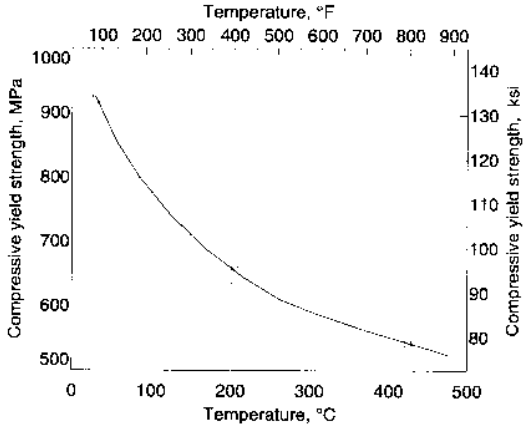
Cast Properties

Ti-6242: Tensile strengths at high temperatures



Consumable electrode skulls melted in water-cooled copper crucibles and cast into tungsten-lined ceramic molds. 6.4 mm (0.250 in.) diam tensile bars machined from 31.7 mm (1.25 in.) thick sections of heat treated, variable thickness step castings. Specimens were treated at 1035 °C (1900 °F), 1 h, HeC + 595 °C (1100 °F), 4 h, HeC.
Source: *Aerospace Structural Metals Handbook*, Vol 4, Code 3718, Battelle Columbus Laboratories, June 1978

Ti-6242: Cast compressive yield strength



Alloy was supplied as cast wedges (tapered plates) of dimension 150 by 165 mm (6 by 6.5 in.). Chemical composition was 6.02 wt% Al, 0.018 wt% C, 0.010 wt% Fe, 0.0047 wt% H, 2.07 wt% Mo, 0.013 wt% N, 0.05 wt% Si, 2.04 wt% Sn and 3.80 wt% Zr. Material was tested in the as-received, as-cast condition with a compressive modulus of about 114 to 117 GPa (16.5 to 17 × 10^{-6} psi) at RT.
Source: O. Deel, "Engineering Data on New Aerospace Structural Materials," AFML-TR-77-198, Battelle Columbus Laboratories, 1977, p 29

Thermal Stability

Much information has been accumulated on the thermal stability of Ti-6242, due to the importance of this property in critical applications. The extent to which room-temperature properties will be affected by previous exposure will depend on the form and condition of the product as well as the time, temperature, and stress level of exposure. Data in the *Aerospace Structural Metals Handbook* (Vol 4, Code 3718) are extensive, although no systematic or firm generalizations can be made at this time. However, the following conclusions are suggested in the *Aerospace Structural Metals Handbook*.

Short-Time Stability (150 h). For sheet in both the duplex and triplex annealed conditions, 150-h exposure at 540 °C (1000 °F), 172 MPa (25 ksi) produced no deleterious effect on subsequent room-temperature tensile properties. This is also true for exposure at lower temperatures and higher stress levels—for example 480 °C (900 °F) at 310 MPa (45 ksi) and 425 °C (800 °F) at 448 MPa (65 ksi). Likewise, 100-h exposure to 540 °C (1000 °F) without load produces no loss in room-temperature sharp notch tensile strength of either duplex or triplex annealed sheet.

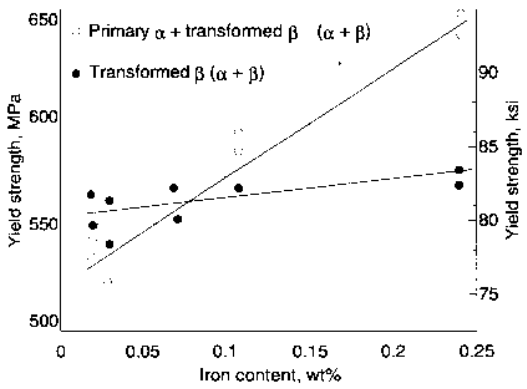
Duplex annealed rolled bar may or may not be affected by exposures of 150 h with load at temperatures up to 540 °C (1000 °F) depending on the exposure stress. At 172 MPa (25 ksi) stress level at 540 °C (1000 °F) no practical loss in room-tem-

High-Temperature Strength

Ti-6242: Typical tensile and compressive properties of sheet at high temperature

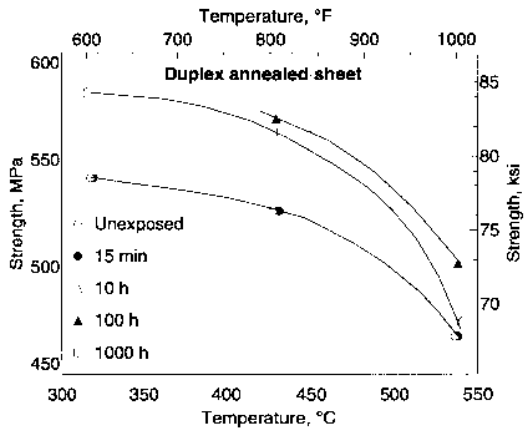
Temperature °C	°F	Ultimate tensile strength MPa	ksi	Tensile yield strength (0.2%) MPa	ksi	Elongation, %	Compressive yield strength (0.2%) MPa	ksi
Longitudinal								
Room temperature		1010	146	990	144	3	1075	156.3
205	400	890	129	760	110	10.8	800	116
370	700	830	120	650	94	11.5	695	101
540	1000	710	103	570	83	17.8	635	92
Transverse								
Room temperature		1020	148	1010	146	2.7	1165	169
205	400	890	129	770	112	10.6	860	125
370	700	830	120	670	97	21.6	750	109
540	1000	720	104	590	86	16.5	695	101

Ti-6242: Hot tensile yield strength of bar at 510 °C vs Fe content



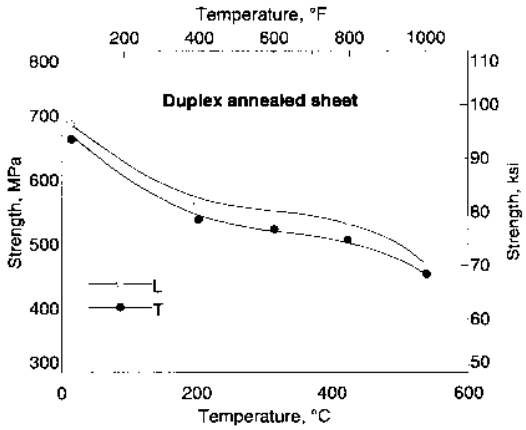
Source: S. Akem and S.R. Seagle, in *Titanium, Science and Technology*, Vol 4, 1985, p 2415

Ti-6242: Ultimate shear strength vs high-temperature exposure



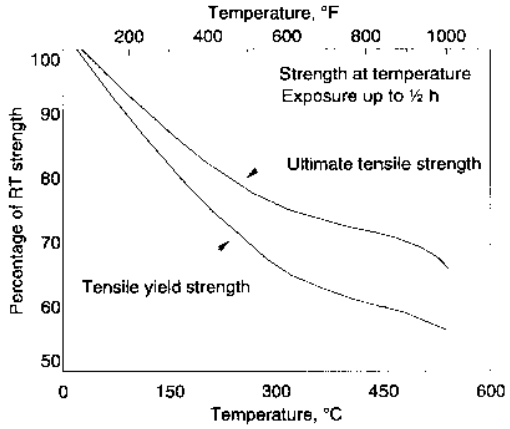
Sheet 1 mm (0.040 in.) thick was heat treated at 900 °C (1650 °F) for 30 min, air cooled, and at 785 °C (1450 °F) for 15 min, air cooled. Material was subsequently exposed to elevated temperatures as indicated. Each data point represents an average of two tests. Longitudinal test results are reported.
Source: C. Dotson, "Mechanical and Thermal Properties of High-Temperature Titanium Alloys," AFML-TR-67-41, 1967, and data from TIMET; reported in *Aerospace Structural Metals Handbook*, Vol 4, Code 3718, Battelle Columbus Laboratories, 1978, p 75

Ti-6242: Ultimate shear strength vs temperature



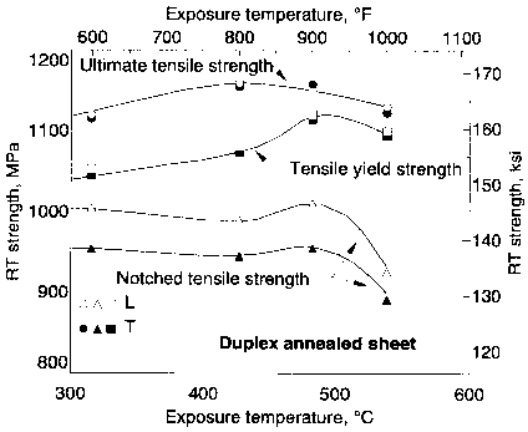
Sheet 1 mm (0.040 in.) thick was heat treated at 900 °C (1650 °F) for 30 min, air cooled, and at 785 °C (1450 °F) for 15 min, air cooled. Each data point represents an average of five tests.
Source: C. Dotson, "Mechanical and Thermal Properties of High-Temperature Titanium Alloys," AFML-TR-67-41, 1967, and data from TIMET; reported in *Aerospace Structural Metals Handbook*, Vol 4, Code 3718, Battelle Columbus Laboratories, 1978, p 75

Ti-6242: Design tensile strength vs temperature



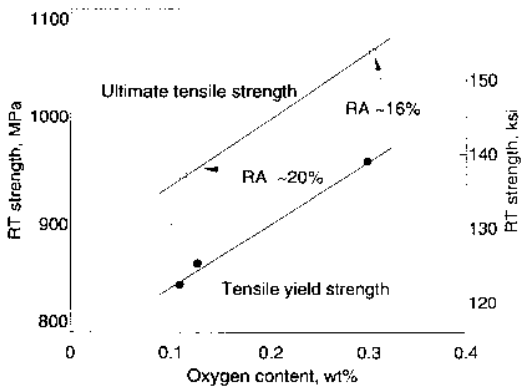
Design allowables for all product forms, duplex- and triplex annealed.
Source: MIL-HDBK 5, 1 Dec 1991

Ti-6242: Effect of 1000-h exposure on RT strength



0.9 mm (0.036 in.) sheet duplex annealed at 900 °C (1650 °F), 30 min, AC + 785 °C (1450 °F), 15 min, AC. RT test results reported. All specimens were acid pickled after exposure prior to tensile tests. Source: *Aerospace Structural Metals Handbook*, Vol 4, Code 3718, Battelle Columbus Laboratories, June 1978

Ti-6242: Effect of oxygen on cast tensile strength



Specimens were consumable electrode skulls melted in water-cooled copper crucibles and cast into tungsten-lined ceramic molds. 6.4 mm (0.250 in.) diam tensile bars cast 0.25 to 0.4 mm (0.010 to 0.015 in.). Oversize and machined to final dimensions following heat treatment of 595 °C (1100 °F), 8 h, AC or HeC. Source: *Aerospace Structural Metals Handbook*, Vol 4, Code 3718, Battelle Columbus Laboratories, June 1978

Cast Tensile Properties

Ti-6242: Room-temperature tensile properties of cast specimens

Heat treatment	Specimen type(a)	Ultimate tensile strength MPa	Ultimate tensile strength ksi	Tensile yield strength MPa	Tensile yield strength ksi	Elongation in 25 mm (1 in.), %	Reduction of area, %	Young's modulus GPa	Young's modulus 10^6 psi
595 °C (1100 °F), 8 h, AC	(b)	948	137.5	853	123.7	10.5	21.4	...	...
		959	139.2	896	130.0	9.0	20.5	...	...
		968	140.4	853	123.7	10.0	18.2	...	...
955 °C (1750 °F), 1 h, AC + 595 °C (1100 °F), 8 h, AC	(b)	1010	146.5	894	129.7	8.0	21.5	127	18.4
		984	142.7	879	127.6	5.0	13.3	118	17.2
		993	144.1	892	129.4	4.0	8.1	117	17.1
1035 °C (1900 °F), 1 h, WQ + 595 °C (1100 °F), 8 h, AC	(b)	1272	184.5	1228	178.2	1.0	3.9	...	...
		1294	187.7	(c)	(c)	1.0	3.9	...	...
		1237	179.4	1237	179.4	1.0	3.1	...	...
1035 °C (1900 °F), 1 h, HeC + 595 °C (1100 °F), 4 h, HeC	(d), (e)	973	141.2	878	127.4	7.8	19.3	...	...
	(d), (f)	957-990	138.8-143.6	862-899	125.0-130.5	5.0-12.0	10.9-24.6	...	...

(a) Specimens were consumable electrode skulls melted in water-cooled copper crucibles and cast into tungsten-lined ceramic molds, unless otherwise indicated. (b) 6.4 mm (0.250 in.) diam tensile bars cast 0.25 to 0.38 mm (0.010 to 0.015 in.) oversize and machined to final dimensions following heat treatment. (c) Failed at less than 0.2% indicated strain. (d) 6.4 mm (0.250 in.) diam tensile bars machined from three thickest sections of heat treated, variable thickness step castings. (e) Average of nine values from three castings. (f) Range of nine values from three castings. Source: *Aerospace Structural Metals Handbook*, Vol 4, Code 3718, Battelle Columbus Laboratories, June 1978

Ti-6242: Typical room-temperature mechanical properties of sheet

Direction	Ultimate tensile strength, MPa	Ultimate tensile strength, ksi	Tensile yield strength (0.2% offset), MPa	Tensile yield strength (0.2% offset), ksi	Elongation, %	Compressive yield strength (0.2%), MPa	Compressive yield strength (0.2%), ksi	Shear strength, MPa	Shear strength, ksi
Longitudinal	1010	146	990	144	13	1075	156.3	690	100
Transverse	1020	148	1010	146	12	1165	169	695	101

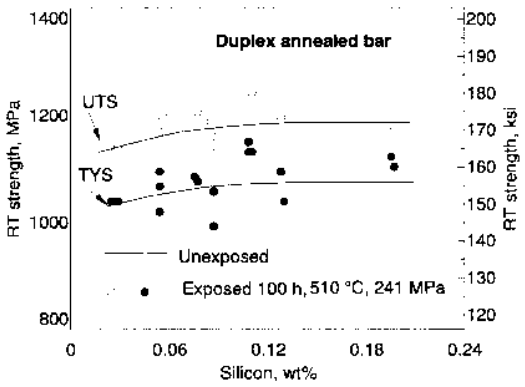
Source: *Metals Handbook*, Vol 3, 9th ed., American Society for Metals, 1987

Ti-6242: Effect of thermomechanical processing on tensile properties

Condition	Tensile yield strength, MPa	Tensile yield strength, ksi	Ultimate tensile strength, MPa	Ultimate tensile strength, ksi	Elongation, %	Reduction of area, %
Ti-6242						
α + β forged + STA(a)	1028	149	938	136	16	35
Ti-6242 + 0.25 wt% Si						
α + β forged + 965 °C (1770 °F), 1 h, AC + 595 °C (1100 °F), 8 h, AC	1007	146	1063	154	13	34
α + β forged + 965 °C (1770 °F), 1 h, OQ + 595 °C (1100 °F) 8 h, AC	1090	158	1187	172	12	20
β forged + 965 °C (1770 °F), 1 h, AC + 595 °C (1100 °F), 8 h, AC	952	138	1035	150	11	25
β forged + 965 °C (1770 °F), 1 h, AC + 595 °C (1100 °F), 8 h, AC	1042	151	1145	166	7	15

(a) STA = 985 °C (1810 °F) (T_β − 15 °C (27 °F)), air cool + 595 °C (1100 °F), 8 h, air cool. Source: M.M. Allen, AFML TR-71-78, Feb 1976

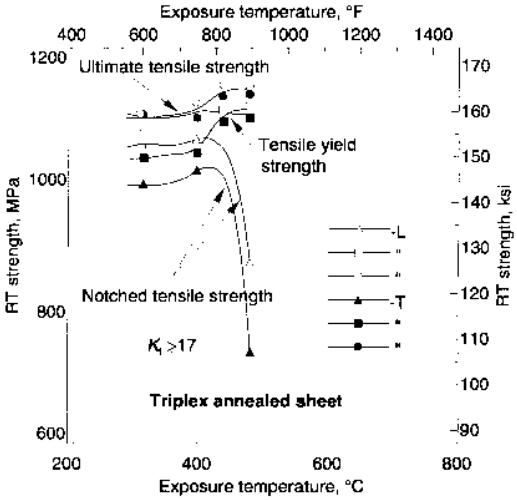
Ti-6242: Effect of silicon and exposure at 510 °C on tensile properties



Specimens were of standard composition except for silicon. 15.8 mm (0.625 in.) bar. Heat treatment: 15 °C (25 °F) below β transus for 1 h, AC, 595 °C (1100 °F) for 8 h, AC.
Source: *Aerospace Structural Metals Handbook*, Vol 4, Code 3718, Battelle Columbus Laboratories, June 1978

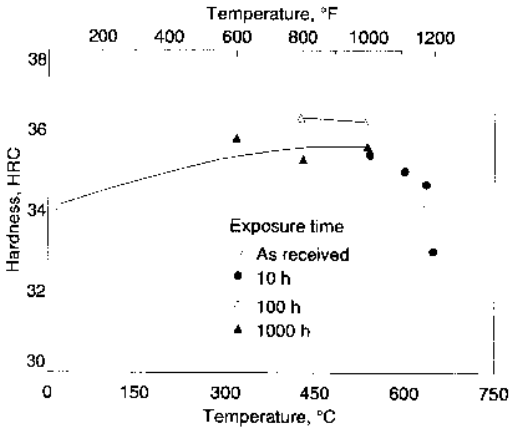
Notched Tensile Strength

Ti-6242: Effect of 1000-h exposure on RT strength



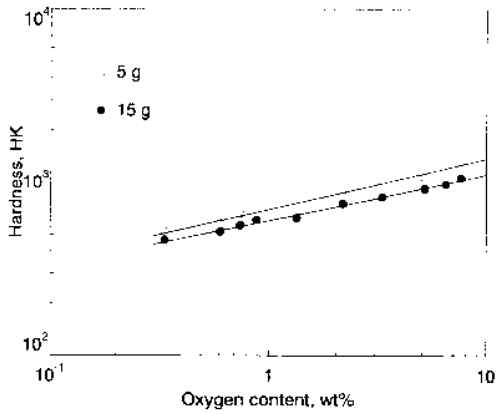
0.9 mm (0.036 in.) sheet triplex annealed at 900 °C (1650 °F), 30 min, AC + 785 °C (1450 °F), 15 min, AC + 595 °C (1100 °F), 2 h, AC. RT test results reported. All specimens were acid pickled after exposure prior to tensile tests.
Source: *Aerospace Structural Metals Handbook*, Vol 4, Code 3718, Battelle Columbus Laboratories, June 1978

Ti-6242: RT hardness vs high-temperature exposure



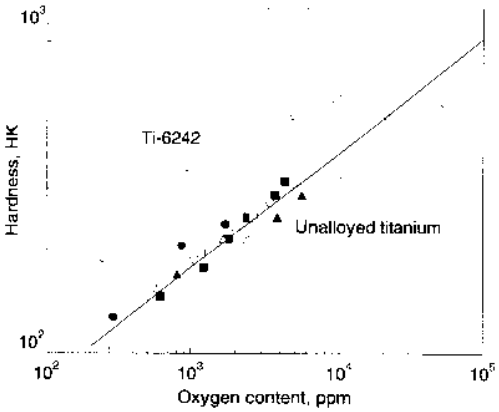
Duplex annealed sheet, 1.0 mm (0.040 in.) thick, heat treated for 30 min at 900 °C (1650 °F) and air cooled, then for 15 min at 790 °C (1450 °F), and air cooled.
Source: C. Dotson, "Mechanical and Thermal Properties of High-Temperature Titanium Alloys," AFML-TR-67-41, Apr 1967; reported in *Aerospace Structural Metals Handbook*, Vol 4, Code 3718, Battelle Columbus Laboratories, 1978, p 27

Ti-6242: Hardness of foil vs oxygen content



Specimens were foil, 80 μm (3.15 mils) thick. Composition was 5.80 wt% Al, 0.02 wt% C, 0.20 wt% Fe, 0.05 wt% Fe, 0.05 wt% H, 2.06 wt% Mo, 0.01 wt% N, 0.06 wt% Si, 1.87 wt% Sn, and 4.00 wt% Zr, <0.4 wt% others. Several foil specimens were oxidized in laboratory air at 855 °C (1570 °F) for various lengths of time and were subsequently homogenized by annealing for 120 h at 870 °C (1600 °F) in vacuum (7×10^{-7} torr). Hardness measurements were made using loads of 5 and 15 g. At least 15 Knoop indentations were made for each data point.
Source: R. Shenoy, J. Unnam, and R. Clark, "Oxidation and Embrittlement of Ti-6Al-2Sn-4Zr-2Mo Alloy," *Oxid. Met.*, Vol 26, 1986, p 105

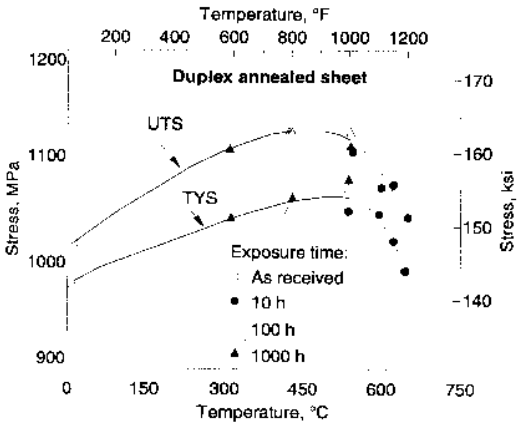
Ti-6242: Hardness vs oxygen content



Bar stock, 28.58 mm (1.125 in.) in diameter. Composition: 6.4 wt% Al, 0.025 wt% C, 0.05 wt% Fe, 0.007 wt% H, 2.0 wt% Mo, 0.006 wt% N, 0.082 wt% O, 2.0 wt% Sn, and 4.2 wt% Zr. Disk specimens were machined to a diameter of 15 mm (0.60 in.) and a thickness of 3.2 mm (0.125 in.), cleaned with methanol, and oxidized for 5 to 1000 h at temperatures from 540 to 870 °C (1000 to 1600 °F). Oxidized specimens were metallographically prepared to present a chord surface with chord length approximately half the disk diameter. Hardness determinations were made with a Kentron microhardness tester using a Knoop indenter under a 1000-g load. Two to four traverses were made on each specimen.
Source: C. Shamblen and T. Redden, "Air Contamination and Embrittlement of Titanium Alloys," in *The Science, Technology and Application of Titanium*, R. Jaffee, Ed., Pergamon Press, 1970, p 199

Tensile Properties

Ti-6242: Effect of exposure on RT tensile strength



Specimens were 1 mm (0.040 in.) sheet duplex annealed at 900 °C (1650 °F), 30 min, AC + 785 °C (1450 °F), 15 min, AC. Longitudinal RT test results are reported. Exposure time as indicated.
Source: *Aerospace Structural Metals Handbook*, Vol 4, Code 3718, Battelle Columbus Laboratories, June 1978

Mechanical Properties

Design Allowables

Ti-6242: Design tensile and compressive strengths of duplex annealed sheet

Basis	Ultimate tensile strength (L-LT)(a) MPa	ksi	Tensile yield strength (L-LT)(a) MPa	ksi	Compressive yield strength (L-LT)(a) MPa	ksi
S-basis	958	139	889-903	129 131(b)	...	...
A-basis(c)	930	135	862	125	910	132
B-basis	986	143	924-937	134-136(b)	979	142

Note: Duplex annealed (DA) sheet thicknesses up to 4.75 mm (0.187 in.) per AMS 4919. (a) Longitudinal (L) and long transverse (LT) properties unless noted. (b) L and LT properties, respectively. (c) 8% elongation minimum for 0.635 to 1.57 mm (0.025 to 0.062 in.) thicknesses and 10% elongation for 1.57 to 4.75 mm (0.062 to 0.187 in.) thicknesses. Source: MIL-HDBK-5, Dec 1991

Ti-6242: Design bearing strengths of duplex annealed sheet

Thickness mm	in.	Basis	Ultimate bearing strength e/D = 1.5 MPa	ksi	e/D = 2.0 MPa	ksi	Bearing yield strength e/D = 1.5 MPa	ksi	e/D = 2.0 MPa	ksi
<1.17	<0.046	A	1344	195	1496	217	1179	171	1392	202
		B	1420	206	1586	230	1261	183	1496	217
1.19-2.36	0.047-0.093	A	1413	205	1675	243	1179	171	1392	202
		B	1496	217	1779	258	1261	183	1496	217
2.38-3.55	0.094-0.14	A	1475	214	1834	266	1179	171	1392	202
		B	1565	227	1944	282	1261	183	1496	217
3.58-4.75	0.141-0.187	A	1510	219	1923	279	1179	171	1392	202
		B	1599	232	2034	295	1261	183	1496	217

Note: Dry pin bearing values of duplex annealed (DA) sheet per AMS 4919. Source: MIL-HDBK-5, Dec 1991

Ti-6242: Design tensile properties of DA bar, forgings, and TA sheet

Product	Basis	Ultimate tensile strength MPa	ksi	Tensile yield strength MPa	ksi	Elonga -tion(a), %	RA(a), %
Bar and forgings ≤75 mm (3 in.) thick							
STA (duplex annealed) bar(b) per AMS 4975	S	896(c)	130(b)	827(c)	120 (c)	10(c)	25(c)
	A	951(d)	138(d)	862(d)	125(d)	...	...
	B	993(d)	144(d)	903(d)	131(d)	...	...
STA (duplex annealed) forgings(e) per AMS 4976	S	896	130	827	120	10(c)	25(c)
Triplex annealed (TA) sheet(f)							
Sheet ≤4.75 mm (0.187 in.) per MIL-T-9046	S	999(g)	145(g)	930(g)	135(g)	(g, h)	...

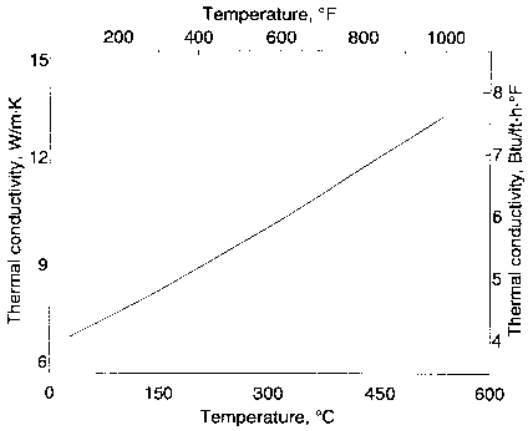
Note: Properties applicable in L, LT, and ST directions, except as noted. (a) S-basis. (b) Cross-sectional area less than 103 cm^2 (16 $in.^2$). (c) Applicable in LT and ST direction if transverse direction is greater than 63.5 mm (2.5 in.). (d) Longitudinal direction only. (e) Cross-sectional area less than 58 cm^2 (9 $in.^2$). (f) S-basis values are representative of test specimens excised from duplex annealed material and then treated to achieve a triplex annealed condition in the laboratory. (f) Specified only in L and LT directions. (g) 8% for 0.64 to 1.57 mm (0.025 to 0.062 in.) and 10% for thickness above 1.57 mm (0.062 in.). Source: MIL-HDBK-5, Dec 1991

Hardness

Typical room-temperature hardness is 32 to 36 HRC for duplex annealed Ti-6242.

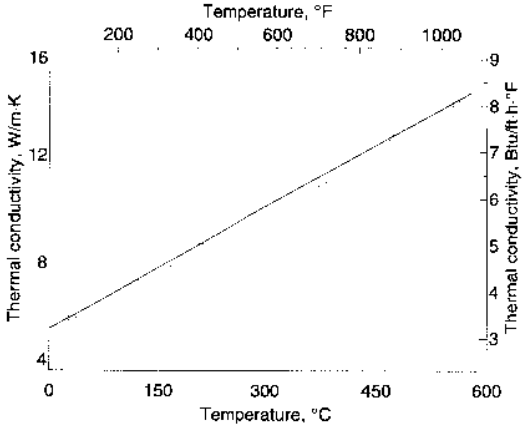
Thermal Conductivity

Ti-6242: Thermal conductivity



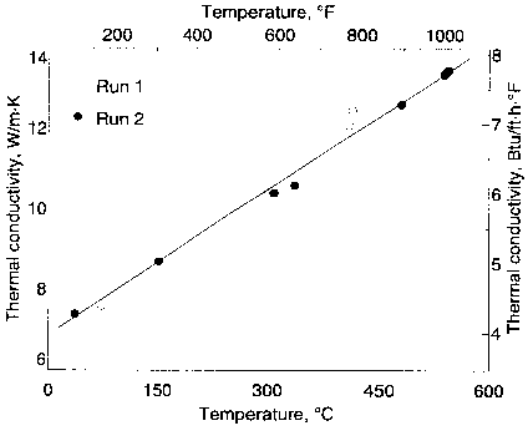
Source: MIL-HDBK-5E, Vol 2, June 1987, p 5.44

Ti-6242: Thermal conductivity of duplex annealed bar



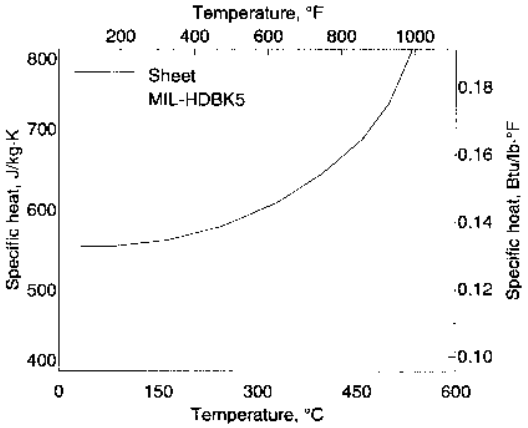
Measurements were taken parallel to the long direction of 54 mm (2⅛ in.) diameter bar. Bar annealed at 900 °C (1650 °F) for 1 h, air cooled, then at 595 °C (1100 °F) for 8 h, air cooled.
Source: H. Russell, "Physical Properties of Ti-6Al-2Sn-4Zr-2Mo," TMCA Case Study M-109, June 1967; reported in *Aerospace Structural Metals Handbook*, Vol 4, Code 3718, Battelle Columbus Laboratories, 1978, p 36,37

Ti-6242: Thermal conductivity of duplex annealed sheet



Measurements were taken parallel to the rolling direction. Alloy used as 1.0 mm (0.040 in.) sheet, annealed at 900 °C (1650 °F) for 30 min, air cooled, then at 790 °C (1450 °F) for 15 min, and air cooled.
Source: C. Dotson, "Mechanical and Thermal Properties of High-Temperature Titanium Alloys," AFML-TR-67-41, Apr 1967

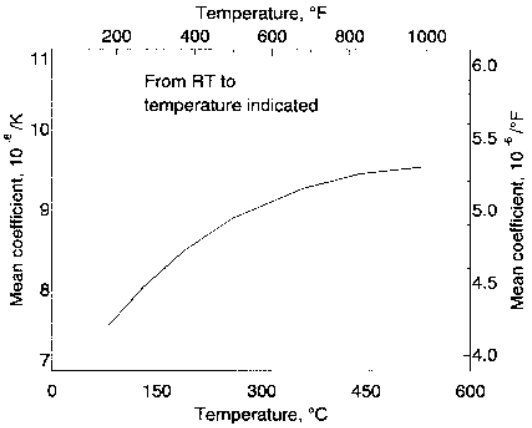
Ti-6242: Specific heat of duplex annealed sheet



1.0 mm (0.040 in.) sheet, annealed at 900 °C (1650 °F) for 30 min, air cooled, then at 790 °C (1450 °F) for 15 min, air cooled.
Source: C. Dotson, "Mechanical and Thermal Properties of High-Temperature Titanium Alloys," AFML-TR-67-41, Apr 1967; reported in *Aerospace Structural Metals Handbook*, Vol 4, Code 3718, Battelle Columbus Laboratories, June 1978

Thermal Expansion

Ti-6242: Thermal coefficient of linear expansion of sheet



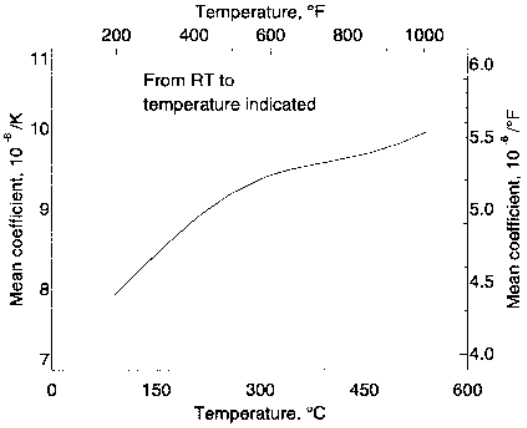
Alloy used as 1.0 mm (0.040 in.) thick sheet, heat treated at 900 °C (1650 °F) for 30 min, then air cooled, and heated at 790 °C (1450 °F) for 15 min, and air cooled. Values are averages of longitudinal and transverse directions.
Source: C. Dotson, "Mechanical and Thermal Properties of High-Temperature Titanium Alloys, AFML-TR-67-41, Apr 1967

Ti-6242: Thermal coefficient of linear expansion

Temperature °C	Temperature °F	Coefficient of thermal expansion 10^{-6}/K	Coefficient of thermal expansion 10^{-6}/°F
0-100	32-212	7.7	4.3
0-315	32-600	8.1	4.5
0-540	32-1000	8.1	4.5

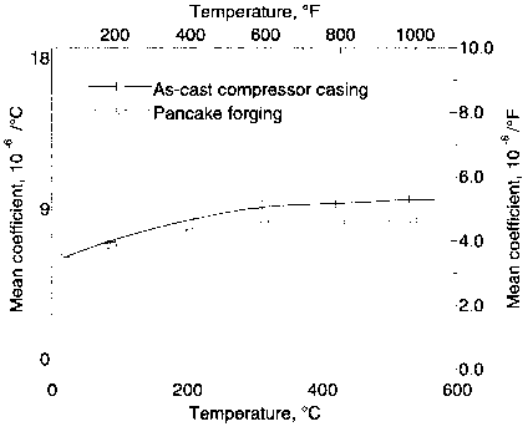
Source: RMI Titanium Basic Design

Ti-6242: Thermal coefficient of linear expansion of bar



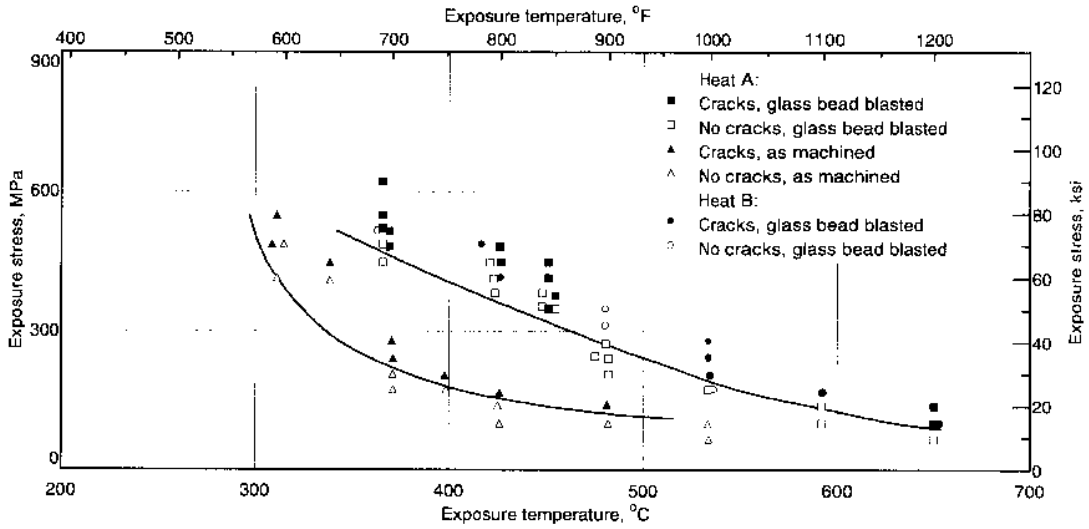
Alloy used as 29 mm (1 1/8 in.) diameter bar, heat treated at 900 °C (1650 °F) for 1 h and air cooled, then at 595 °C (1100 °F) for 8 h, and air cooled. Measurements were taken in the longitudinal direction; values are averages of ascending and descending temperature values. Source: H. Russell, "Physical Properties of Ti-6Al-2Sn-4Zr-2Mo," TMCA Case Study M-109, June 1967

Ti-6242: Thermal coefficient of linear expansion of various product forms



Specimens were an as-cast compressor casing and pancake forgings heat treated at 955 °C (1750 °F) for 1 h and air cooled, then at 595 °C (1100 °F) for 8 h and air cooled.
Source: *Aerospace Structural Metals Handbook*, Vol 4, Code 3718, Battelle Columbus Laboratories, 1978, p 37

Ti-6242: 100-h hot-salt SCC



Hot-salt stress-corrosion cracking characteristics of duplex annealed compressor disc forging. Heat A: 975 °C (1790 °F), 1 h, AC + 595 °C (1100 °F), 8 h, AC; Heat B: 950 °C (1750 °F), 1 h, AC + 595 °C (1100 °F), 8 h, AC.
Source: *Aerospace Structural Metals Handbook*, Vol 4, Code 3718, June 1978, p 43

Thermal Properties

Melting Point. Approximately 1700 °C (3100 °F)

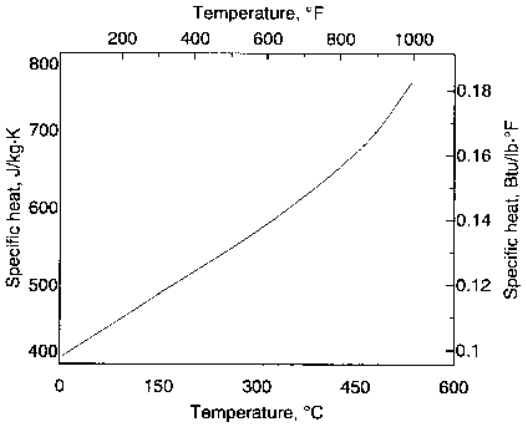
Thermal Coefficient of Linear Expansion. 7.7 μm/m · K (4.3 μin./in. · °F) at 205 °C (400 °F); 8.1 μm/m · K (4.5 μin./in. · °F) at 315 to 540 °C (600 to 1000 °F)

Specific Heat. 460 J/kg · K (0.110 Btu/lb · °F) at 100 °C (212 °F)

Thermal Conductivity. 7.1 W/m · K (4.1 Btu/ft · h · °F) at 100 °C (212 °F)

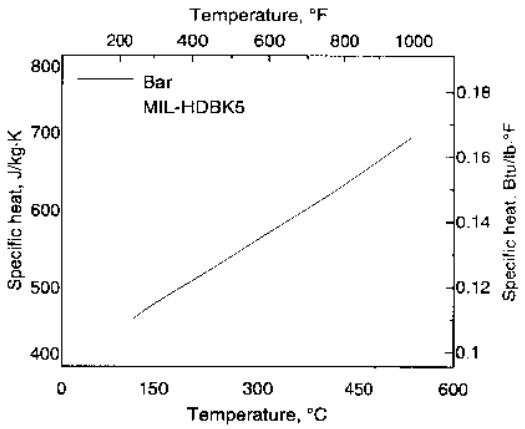
Specific Heat

Ti-6242: Specific heat vs temperature



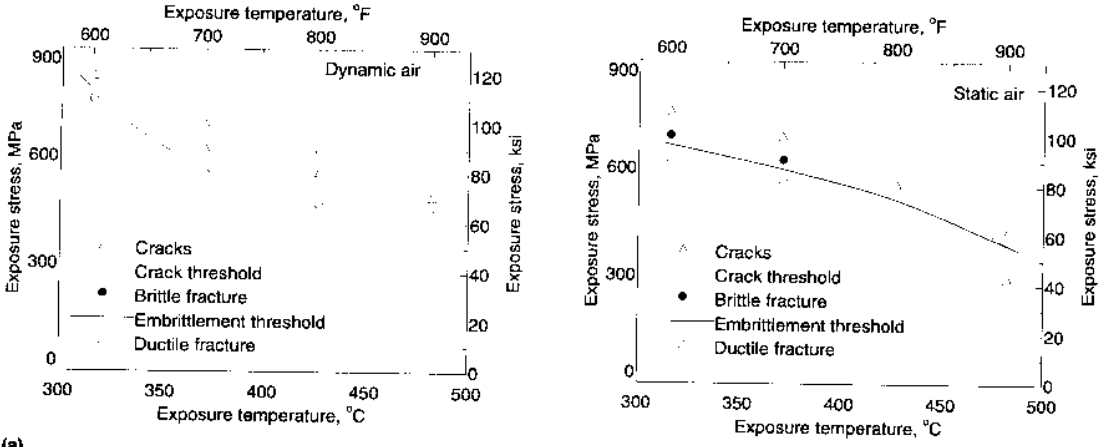
Source: MIL-HDBK-5E, Vol 2, June 1987, p 5.44

Ti-6242: Specific heat of duplex annealed bar



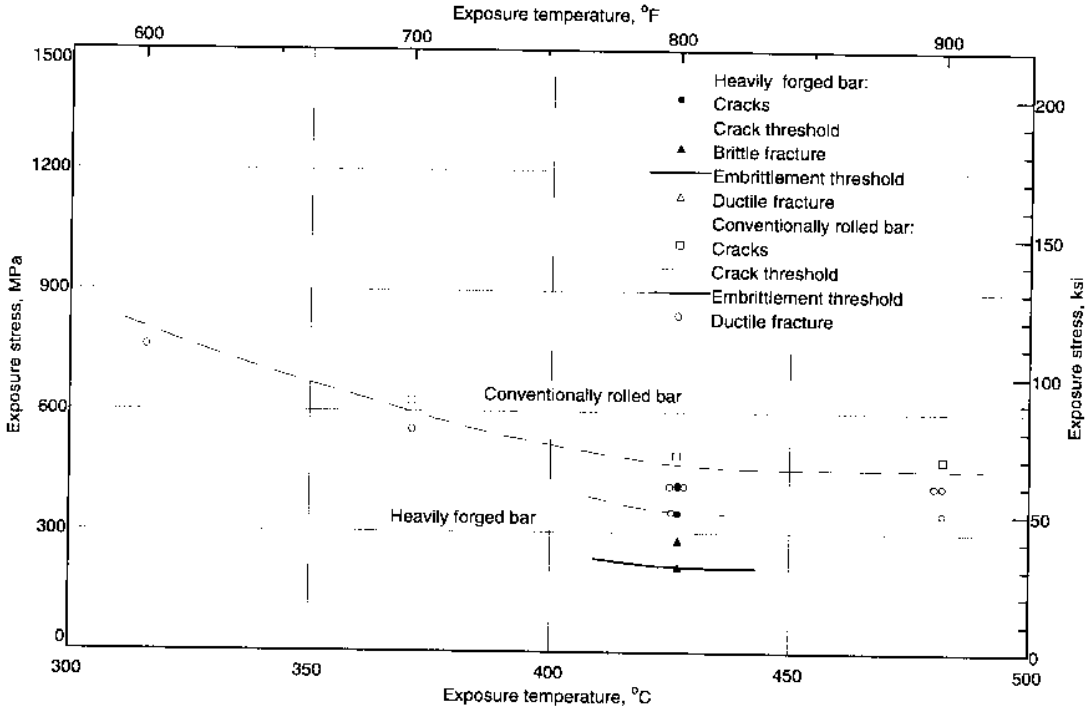
29 mm (1⅛ in.) diameter bar, annealed at 900 °C (1650 °F) for 1 h, air cooled, then at 595 °C (1100 °F) for 8 h, air cooled.
Source: H. Russell, "Physical Properties of Ti-6Al-2Sn-4Zr-2Mo," TMCA Case Study M-109, June 1967; reported in *Aerospace Structural Metals Handbook*, Vol 4, Code 3718, Battelle Columbus Laboratories, 1978, p 37

Ti-6242: 100-h crack thresholds for NaCl-coated bar



Little or no effect is noted between (a) dynamic air and (b) static air environment.
100-h embrittlement and crack threshold curves for NaCl-coated (aerosol deposited on chemically milled tubular specimens) (from mill annealed bar) exposed to elevated temperature with load in either a laboratory or simulated aircraft turbine engine compressor environment and subsequently tensile tested at room temperature.
Source: H.R. Gray, "Relative Susceptibility of Titanium Alloys to Hot-Salt Stress-Corrosion," NASA TN D-6498, Nov 1971

Ti-6242: Crack threshold of heavily forged bar



Effect of alloy processing on 100-h embrittlement and crack threshold curves for NaCl-coated (aerosol deposited on hot surface) chemically milled tubular specimens (from duplex annealed bar) exposed to elevated temperature with load in a laboratory environment and subsequently tensile tested at room temperature.
Source: *Aerospace Structural Metals Handbook*, Vol 4, Code 3718, June 1978, p 40

Ti-6242: Precracked tensile strength in air and in 3.5% NaCl
Center fatigue cracked samples

Sheet gage			Air (continuously loaded)		Notch tensile strength in: Saltwater (sustained load) Failure in 1 to 2 h		Failure in <1 h	
mm	in.	Direction	MPa	ksi	MPa	ksi	MPa	ksi
Duplex anneal(a)								
11.4	0.45	T	...	...	758(c)	110(c)	(d)	(d)
2.0	0.080	L	875	127	696	101	(d)	(d)
2.0	0.080	T	827	120	696	101	744	108
3.1	0.125	L	889	129	765	111	(d)	(d)
3.1	0.125	T	827	120	696	101	744	108
Triplex anneal(b)								
0.4	0.015	T	...	...	689	100	(d)	(d)
2.0	0.080	L	882	128	...	...	434(e)	63(e)
2.0	0.080	T	855	124	600	87	(d)	(d)
3.1	0.125	L	827	120	496	72	551	80
3.1	0.125	T	744	108	...	...	365	53

(a) 900 °C (1650 °C), 30 min, AC + 785 °C (1450 °F), 15 min, AC. (b) 900 °C (1650 °F), 30 min, AC + 785 °C (1450 °F), 15 min, AC + 595 °C (1100 °F), 2 h, AC. (c) No failure in 68 h. (d) Failure stress not established. (e) Failed in 5 min. Source: R. Wood and R. Favor, *Titanium Alloys Handbook*, MCIC-HDBK-02, Battelle, 1972

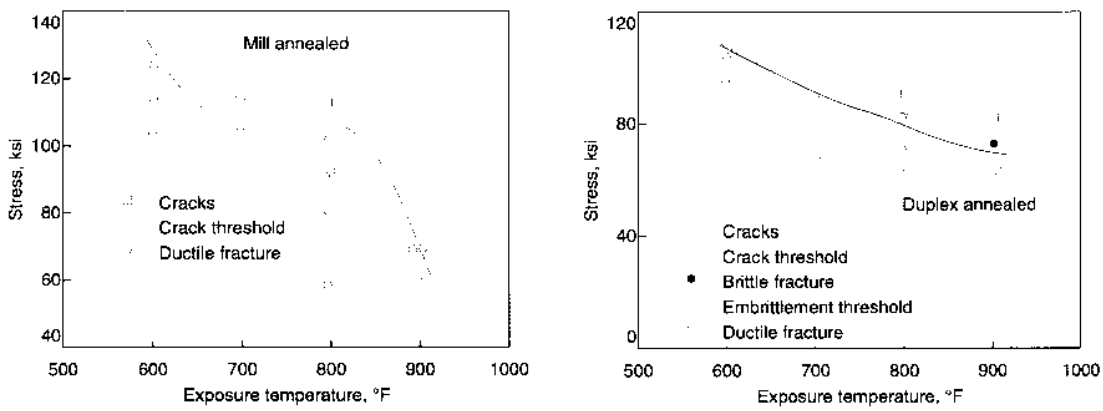
Hot-Salt SCC Factors

Hot-salt SCC behavior of Ti-6242 in laboratory tests is influenced by several factors. Oxygen is necessary for hot-salt cracking to occur. At least one study has shown that cracking will not occur in the highly susceptible Ti-5Al-2.5Sn when the environmental pressure is reduced below 10 μm. Although the role of water (moisture) has not been clearly established, it appears that water is also a necessary environmental component in the cracking process (*Metals Handbook, Corrosion*, Vol 13, 9th ed., 1987, p 689).

Air velocity appears to have no effect on Ti-6242 cracking (see figure) and only a minor effect on Ti-8Al-1Mo-1V cracking (H.R. Gray and J.R. Johnston,"Hot-Salt Stress-Corrosion of a Titanium Alloy Under Simulated Turbine-Engine Compressor Environment," NASA Technical Note, TND-5510, Oct 1969).

Effect of Processing. There appears to be little difference between mill annealing and duplex annealing of Ti-6242 bar (see figure). Heavy forging can reduce the embrittlement and cracking thresholds. The cracking thresholds reported for sheet at 100-h exposure are generally lower than those reported for bar.

Ti-6242: 100-h crack threshold of NaCl-coated bar



Little or no effect is noted between (a) mill annealed bar and (b) duplex annealed bar.
5 100-h embrittlement and crack threshold curves for NaCl-coated (aerosol deposit on hot surface) on machined tubular specimens exposed to elevated temperature with load in simulated aircraft turbine engine compressor environment and subsequently tensile tested at room temperature. Source: H.R. Gray, NASA TN D-6498, Nov 1971

(the so-called accelerated crack-growth type of salt-stress corrosion) and at elevated temperatures in the presence of a halogen salt (e.g., NaCl). The SCC susceptibility of Ti-6242 in hot salt appears to be less than that of Ti-8Al-1Mo-1V and Ti-6Al-4V (see figure). In ambient salt solution, the SCC threshold of Ti-6242 in the STA condition is comparable to mill annealed (MA) Ti-8Al-1Mo-1V (see table).

A substantial amount of laboratory testing on the hot-salt SCC behavior of Ti-6242 has been performed, but like other susceptible alloys (such as Ti-8Al-1Mo-1V), no failure in service has been attributed to hot-salt cracking. The likely reason for this is the critical relationship between stress and environment, which needs to peak simultaneously for extended periods of time for cracking to occur. Actual engine environmental conditions are unique and considered less conducive to stress corrosion than laboratory exposure conditions. Possible ameliorating engine conditions are high air velocities, high pressures, salt-air conditions, oil contamination, and/or unique operating cycles.

Triplex annealing appears to improve resistance to cracking slightly versus a duplex anneal.

Ambient Salt Solution Cracking

Ti-6242: Comparative toughness in air and stress-corrosion threshold in 3.5% NaCl solution at 25 °C

Alloy	Thickness mm	Thickness in.	Heat treatment	Tensile yield strength MPa	Tensile yield strength ksi	K_{Ic} or K_c MPa√m	K_{Ic} or K_c ksi√in.	K_{Iscc} or K_{scc} MPa√m	K_{Iscc} or K_{scc} ksi√in.
Ti-6Al-2Sn-4Zr-2Mo	13	0.50	STA	1048	152	58	53	29	27
Ti-6Al-2Sn-4Zr-6Mo	13	0.50	MA	1103	160	60	55	22	20
			DA	1034	150	88	80	49	45
	7	0.30	MA	965	140	57	52	28	26
			β STA	1172	170	89	81	49	45
Ti-8Al-1Mo-1V	1.3	0.05	MA	999	145	82	75	33	30
	1.3	0.05	DA	930	135	176	160	55	50
	13	0.5	MA	999	145	52	48	22	20
	13	0.5	DA	930	135	110	100	35	32
	13	0.5	MA,WQ	841	122	>110	>100	46	42
	13	0.5	βST, WQ	868	126	>110	>100	>110	>100
Ti-6Al-4V (standard grade)	13	0.50	MA	944	137	66	60	38	35
			DA	917	133	77	70	57	52
			STA	1103	160	51	47	27	25
			β STA	1068	155	77	70	49	45

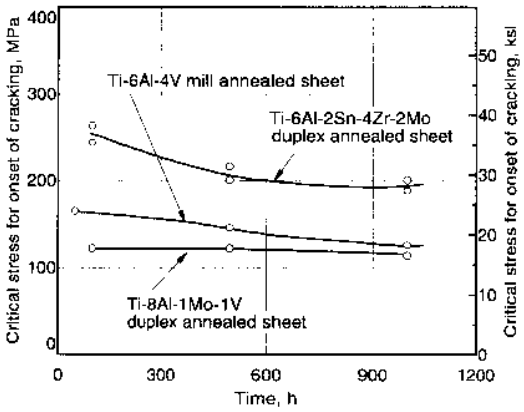
Note: The data were generated in ambient neutral 3.5% NaCl solution. It should be cautioned that these K_{Iscc} values are highly dependent on alloy composition, metallurgical condition, and product form and thickness. Therefore, they may or may not be representative of alloy product materials commercially available. Source: R. Schutz, Stress-Corrosion Cracking of Titanium Alloys, in *Stress-Corrosion Cracking: Materials Performance and Evaluation*, ASM International, 1992

Ti-6242: Effect of heat treatment on saltwater SCC of sheet (precracked)

Heat treatment(a)	Applied net stress MPa	Applied net stress ksi	Time to rupture in 3.5% NaCl
730 °C (1350 °F), 8 h, FC	617	89.5	Broke on loading
	466	67.7	1 min
	308	44.7	7 min
	551	80	>73 h(a)
900 °C (1650 °F),	586	85	>42 h(a)
30 min, AC + 785 °C	689	100	>43h(a)
(1450 °F), 15 min, AC	758	110	>68 h(a)
	551	80	>48 h(a)
900 °C (1650 °F),	586	85	>42 h(a)
30 min, AC + 785 °C	620	90	>24 h(a)
(1450 °F), 15 min, AC + 595 °C (1100 °F), 8 h, AC	689	100	>2 h(a)

(a) No failure, test discontinued. Source: R. Wood and R. Favor, *Titanium Alloys Handbook*, MCIC-HDBK-02, Battelle, 1972

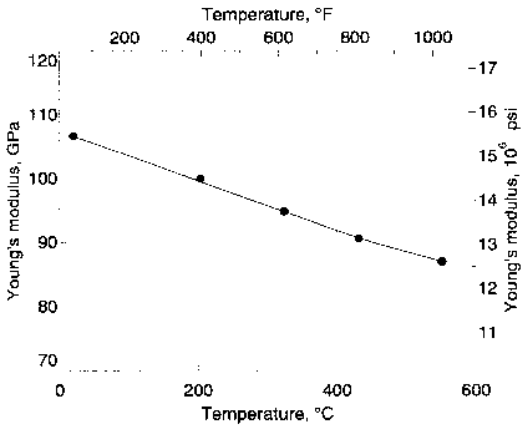
Ti-6242: Comparative critical stress in 425 °C salt



Crack-no crack limit stress.
Source: R. Wood and R. Favor, *Titanium Alloys Handbook*, MCIC-HDBK-02, Battelle, 1972

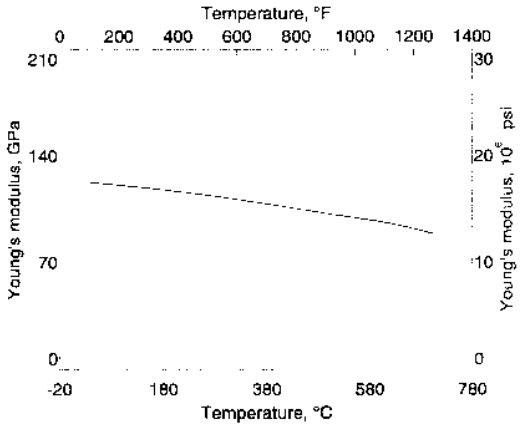
Dynamic Modulus

Ti-6242: Dynamic Young's modulus (tensile)



Specimens were 1 mm (0.040 in.) sheet duplex annealed at 900 °C (1650 °C), 30 min, AC + 790 °C (1450 °F), 15 min, AC. Average of longitudinal and transverse data.
Source: *Aerospace Structural Metals Handbook*, Vol 4, Code 3718, Battelle Columbus Laboratories, 1978, p 85

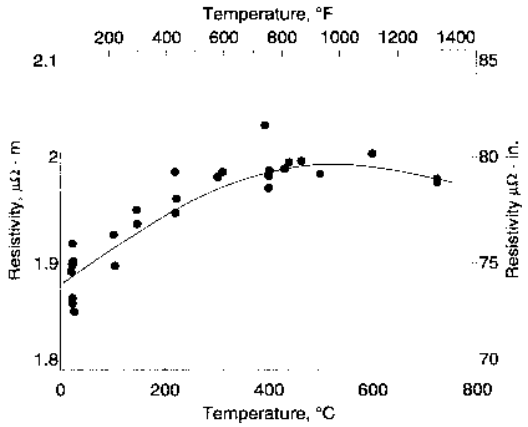
Ti-6242: Dynamic Young's modulus (tensile)



Specimens were duplex annealed pancake forgings, 955 °C (1750 °F), 1 h, AC + 595 °C (1100 °F), 8 h, AC.
Source: *Aerospace Structural Metals Handbook*, Vol 4, Code 3718, Battelle Columbus Laboratories, 1978, p 85

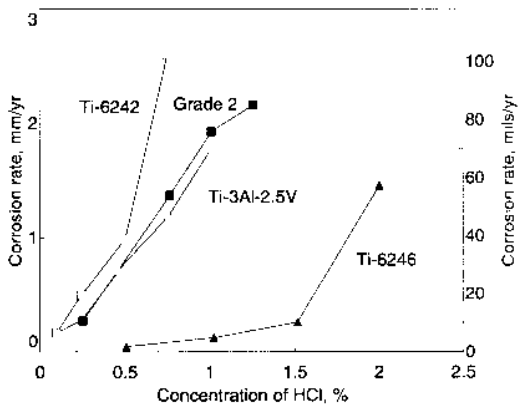
Electrical Resistivity

Ti-6242: Resistivity of duplex-annealed sheet



Specimens were 0.95 mm (0.038 in.) sheet, two heats, two specimens each heat, duplex annealed at 900 °C (1650 °F), 30 min, AC + 785 °C (1450 °F), 15 min, AC. Measurements were made parallel to the rolling direction.
Source: *Aerospace Structural Metals Handbook*, Vol 4, Code 3718, Battelle Columbus Laboratories, 1978, p 38

Ti-6242: Comparative corrosion in boiling HCl



The higher molybdenum content of Ti-6246 increases resistance in reducing environments compared to Ti-6242. General corrosion of annealed titanium alloys in naturally aerated HCl solutions.
Source: R. Schutz, "Corrosion of Titanium and Titanium Alloys," in *Metals Handbook, Corrosion*, Vol 13, 9th ed., 1987, p 680

General Corrosion

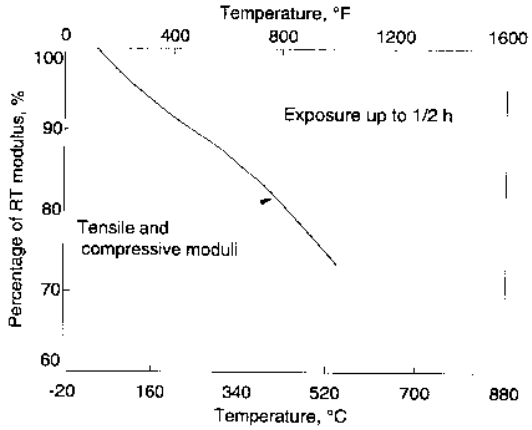
Resistance in Reducing Environments. The corrosion resistance of Ti-6242 in various media is not well documented, although it is probably comparable to other Ti-6Al-base alloys such as Ti-6Al-4V. The molybdenum content of Ti-6242 is not high enough to impart additional corrosion resistance in reducing environments (see figure above). The crevice corrosion resistance of Ti-6242 is probably less than grade 2 titanium, because crevice corrosion generally is associated with acidification from oxidant depletion.

Oxidation. A blue oxide film typically forms in about 6 to 10 h in exposures not exceeding 540 °C (1000 °F). Degradation of mechanical properties from oxidation at longer times and usual service temperatures has not been observed. In a strong oxidizing environment, resistance is probably comparable to grade 2 titanium or Ti-6Al-4V.

Stress-Corrosion Cracking

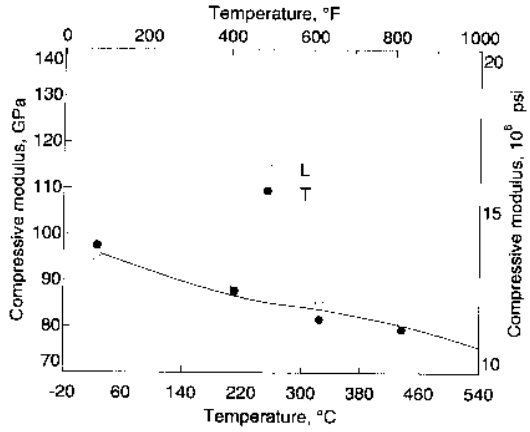
Under stress, Ti-6Al-2Sn-4Zr-2Mo has been shown to be subject to stress-corrosion cracking (SCC)—at room temperature in the presence of aqueous chloride solution and a preexisting crack

Ti-6242: Effect of temperature on moduli



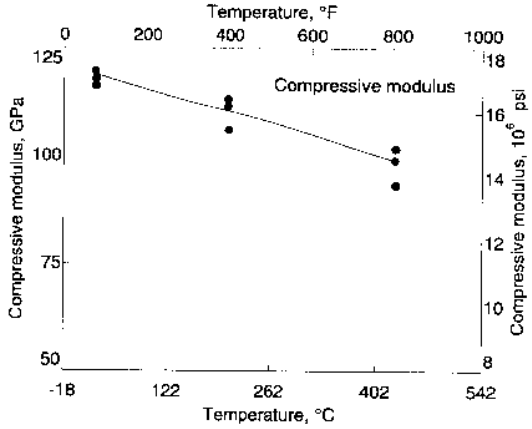
All product forms, duplex and triplex annealed.
Source: MIL-HDBK-5

Ti-6242: Static compressive modulus



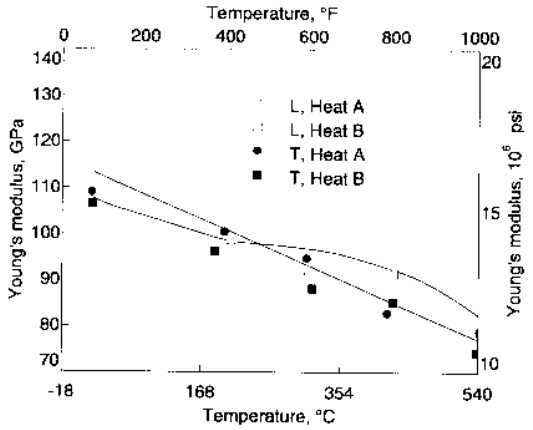
Specimens were 1 mm (0.040 in.) sheet duplex annealed at 900 °C (1650 °F), 30 min, AC + 790 °C (1450 °F), 15 min, AC. Each data point is an average of five tests.
Source: *Aerospace Structural Metals Handbook*, Code 3718, Vol 4, Battelle Columbus Laboratories, 1978, p 85

Ti-6242: Cast compressive elastic modulus



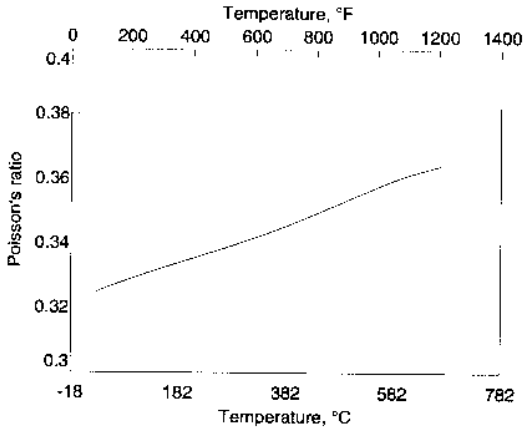
Alloy was supplied as cast wedges (tapered plate) 150 by 165 mm (6 by 6.5 in.). Chemical composition was 6.02 wt% Al, 0.018 wt% C, 0.010 wt% Fe, 0.0047 wt% H, 2.07 wt% Mo, 0.013 wt% N, 0.168 wt% O, 0.05 wt% Si, 2.04 wt% Sn, and 3.80 wt% Zr. Material was tested in the as-received, as-cast condition.
Source: O. Deel, "Engineering Data on New Aerospace Structural Materials," AFML-TR-77-198, Battelle Columbus Laboratories, 1977, p 29

Ti-6242: Static Young's modulus (tensile)



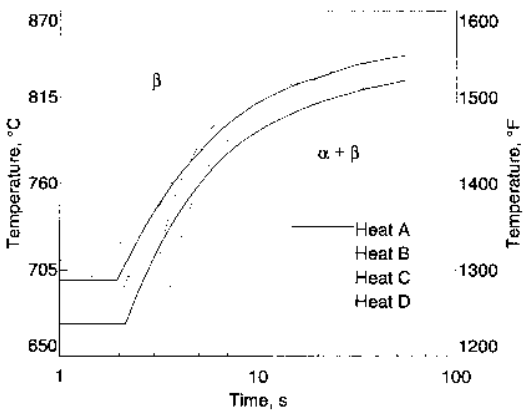
Specimens were 1 mm (0.040 in.) sheet, 30 min, duplex annealed 900 °C (1650 °F), ½ h, AC + 790 °C (1450 °F), 15 min, AC. Each point is an average of ten tests.
Source: *Aerospace Structural Metals Handbook*, Vol 4, Code 3718, Battelle Columbus Laboratories, 1978, p 85

Ti-6242: Poisson's ratio



Specimens were pancake forgings, duplex annealed at 955 °C (1750 °F), 1 h, AC + 595 °C (1100 °F), 8 h, AC.
Source: *Aerospace Structural Metals Handbook*, Vol 4, Code 3718, Battelle Columbus Laboratories, 1978, p 85

Ti-6242: Continuous cooling transformation diagram



Effect of heat-to-heat variations, within specification, on continuous cooling transformation.
Source: D.R. Mitchell, Welding Evaluation of Ti-6Al-2Sn-4Zr-2Mo Sheet, TMCA Project BW-10-1, Final Report, June 1968, reported in *Aerospace Structural Metals Handbook*, Battelle Columbus Laboratories, June 1978

Physical Properties

Ti-6242: Summary of typical physical properties

Beta transus	995 ± 15 °C (1825 ± 25 °F)
Melting (liquidus) point	~1705 °C (3100 °F)
Density(a)	4.54 g/cm^3 (0.164 lb/in.^{3})
Electrical resistivity(a)	1.85 to 1.9 μΩ · m
Magnetic permeability	Nonmagnetic
Specific heat capacity(a)	460 J/kg · K (0.11 Btu/lb · °F)
Thermal conductivity(a)	7 W/m · K (4 Btu/ft · h · °F)
Thermal coefficient of linear expansion(b)	7.7×10^{-6}/°C (4.3×10^{-6}/°F)

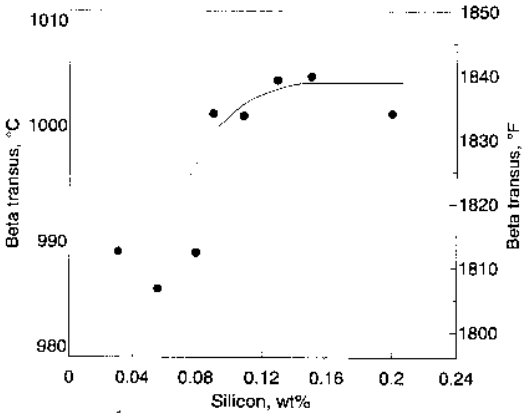
(a) Typical values at room temperature of about 20 to 25 °C (68 to 78 °F). (b) Mean coefficient from 0 to 100 °C (32 to 212 °F)

Elastic Properties (Static)

Ti-6242: As-cast compressive modulus

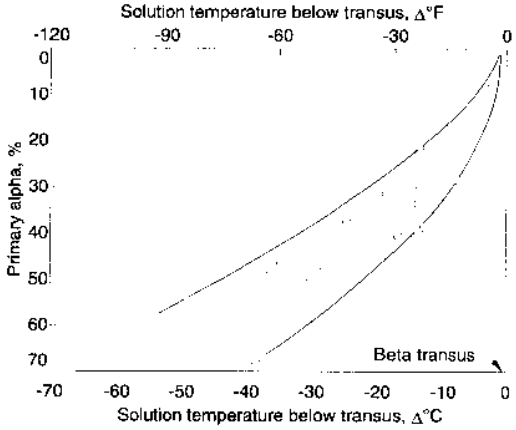
Specimen No.	Tensile yield strength (0.2% offset) MPa	ksi	Compressive modulus GPa	10^6 psi
At room temperature				
1	993	144.3	119	17.3
2	898	130.3	118	17.1
3	910	132.1	116	16.9
At 200 °C (400 °F)				
4	675	98.0	107	15.5
5	615	89.3	114	16.4
6	670	97.2	112	16.3
At 425 °C (800 °F)				
7	546	79.2	103	14.9
8	520	75.4	101	14.7
9	535	77.6	96	13.9

Ti-6242: Effect of silicon on beta transus temperature



Source: *Aerospace Structural Metals Handbook*, Code 3718, Vol 4, Battelle Columbus Laboratories, June 1978

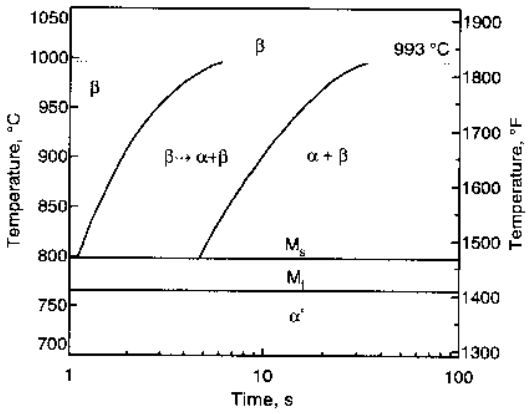
Ti-6242S: Alpha content vs solution temperature



Solution treated, 1 h, AC + 595 °C (1100 °F), 8 h, AC.
Source: *Aerospace Structural Metals Handbook*, Code 3718, Vol 4, Battelle Columbus Laboratories, June 1978

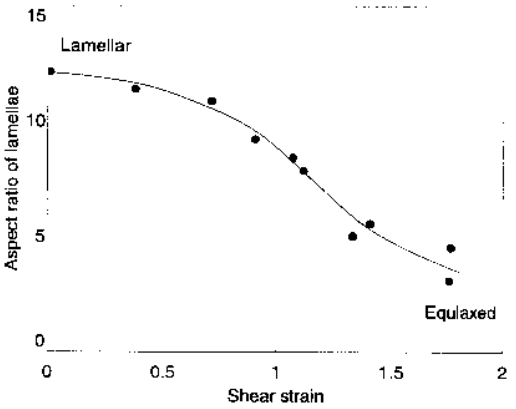
Transformation Kinetics

Ti-6242: Time-temperature-transformation diagram



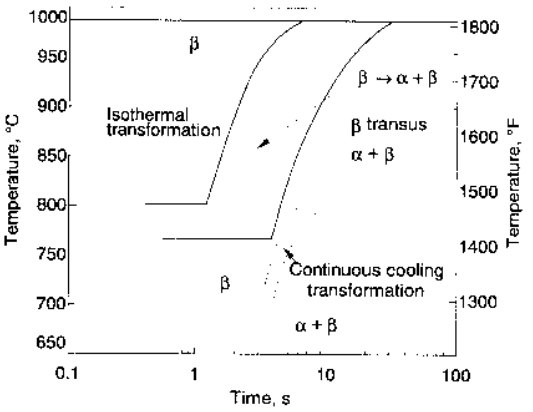
Source: H.W. Rosenberg, M.B. Vordahl, and D.B. Hunter, Project No. 48-8, Tech. Rep. No. 17, Timet, Jan 1966

Ti-6242: Breakup of lamellar α by shear strain



Lamellar breakup was done by torsional deformation at 925 °C (1700 °F), and the transformation from lamellar to equiaxed grains was evaluated by optical and electron microscopy. The average length-to-width aspect ratio of groups of 100 random lamella segments or equiaxed grains was calculated.
Source: G. Welsch, I. Weiss, D. Eylon, and F.H. Froes, Shear Deformation and Breakup of Lamellar Morphology in Ti-6Al-2Sn-4Zr-2Mo, in *Sixth World Conference on Titanium*, P. Lacombe, R. Tricot, and G. Beranger, Ed., Les Editions de Physique, Paris, 1989, p 1289-1293

Ti-6242: Comparison of CCT diagram and isothermal diagram



Source: D.R. Mitchell, Welding Evaluation of Ti-6Al-2Sn-4Zr-2Mo Sheet, TMCA Project BW-10-1, Final Report, June 1968, reported in *Aerospace Structural Metals Handbook*, Battelle Columbus Laboratories, June 1978

Ti-6Al-2Sn-4Zr-2Mo-0.08Si: Specifications and compositions (continued)

Specification	Designation	Description	Composition, wt% Al	Fe	H	Mo	N	O	Sn	Zr	Other
USA (continued)											
AMS 4976D		Frg HT	5.5-6.5	0.1 max	0.0125 max	1.8-2.2	0.05 max	0.15 max	1.8-2.2	3.6-4.4	C 0.05 max; Si 0.06-0.1; Y 0.005 max; OE 0.1 max; OT 0.3 max; bal Ti
MIL T-81556A	Code AB-4	Ext Bar Shp Ann	5.5-6.5	0.25	0.015	1.8 2.2	0.04	0.15	1.8-2.2	3.6-4.4	C 0.05; Y 0.005; Si 0.06-0.1; OT 0.3; bal Ti
MIL T-81556A	Code AB-4	Ext Bar Shp STA	5.5-6.5	0.25	0.015	1.8-2.2	0.04	0.15	1.8-2.2	3.6-4.4	C 0.05; Si 0.06-0.1; Y 0.005; OT 0.3; bal Ti
MIL T-81915	Type III Comp B	Cast Ann	5.5-6.5	0.35	0.015	1.5-2.5	0.05	0.12	1.5-2.5	3.6-4.4	C 0.08; OT 0.4; bal Ti
MIL T-9046J	Code AB-4	Sh Strp Plt DA	5.5-6.5	0.25	0.015	1.8 2.2	0.04	0.15	1.8-2.2	3.6-4.4	C 0.05; OT 0.3; bal Ti
MIL T-9046J	Code AB-4	Sh Strp Plt TA	5.5-6.5	0.25	0.015	1.8-2.2	0.04	0.15	1.8 2.2	3.6-4.4	C 0.05; OT 0.3; bal Ti
MIL T-9047G	Ti-6Al-2Sn-4Zr-2Mo	Bar Bil DA	5.5-6.5	0.25	0.015	1.8-2.2	0.04	0.15	1.8-2.2	3.6-4.4	C 0.05; OT 0.3; Y 0.005; bal Ti
MIL T-9047G	Ti-6Al-2Sn-4Zr-2Mo	Bar Bil STA	5.5-6.5	0.25	0.015	1.8-2.2	0.04	0.15	1.8-2.2	3.6-4.4	C 0.05; Y 0.005; OT 0.3; bal Ti

Ti-6Al-2Sn-4Zr-2Mo-0.08Si : Compositions

Specification	Designation	Description	Composition, wt% Al	Fe	H	Mo	N	O	Sn	Zr	Other
France											
Ugine	UT6242	Bar Frg Ann	5.5-6.5			1.8-2.2			1.8-2.2	3.6-4.4	
Germany											
Deutsche T	Contimet AlSnZrMo 6-2-4-2	Plt Bar Frg Ann	5.5-6.5	0.25	0.015	1.8-2.2	0.05	0.15	1.8-2.2	3.6-4.4	C 0.05; Si 0.06-0.12; bal Ti
Deutsche T	Contimet AlSnZrMo 6-2-4-2	Plt Bar Frg STA	5.5-6.5	0.25	0.015	1.8-2.2	0.05	0.15	1.8-2.2	3.6-4.4	C 0.05; Si 0.06-0.12; bal Ti
Deutsche T	LT 24	Aged	5.5-6.5	0.25	0.015	1.8-2.2	0.05	0.12	1.8-2.2	3.6-4.4	C 0.05; bal Ti
Fuchs	TL62	Frg	6			2			2	4	bal Ti
Japan											
Kobe	KS6-2-4-2	Bar Frg STA	5.5-6.5	0.25	0.015	1.8-2.2	0.05	0.15	1.8-2.2	3.6-4.4	bal Ti
USA											
OREMET	Ti-6242										
RMI	RMI 6Al-2Sn-4Zr-2Mo-0.10Si	Bar Bil Plt Sh STA	5.5-6.5	0.25	0.01-0.0125	1.75-2.25	0.05	0.12	1.75-2.25	3.5-4.5	C 0.08; Si 0.1; bal Ti
TIMET	TIMETAL 6-2-4-2	Bar Bil Plt Sh STA	5.5-6.5	0.25	0.01-0.0125	1.75-2.25	0.05	0.12	1.75-2.25	3.5-4.5	C 0.08; Si 0.1; bal Ti

Phases and Structures

The structures of Ti-6Al-2Sn-4Zr-2Mo alloy are typically equiaxed α in a transformed β matrix, or a fully transformed structure that maximizes creep resistance. The equiaxed α grains found in sheet products tend to be smaller than those found in forgings, as with other alloys, and are present in greater proportion than in forgings. Primary α is typically about 80 to 90% of the structure in sheet products and can be somewhat lower than this in forged products, because the final forging temperature is normally higher than the final rolling temperature used for sheet. As in other near-α alloys, small amounts of residual β phase can be observed metallographically within the transformed β portion of the structure, typically between the acicular α grains of the transformed phase. Breakup of lamellar α into equiaxed α occurs during working (see figure).

Beta Transus. 995 ± 15 °C (1825 ± 25 °F)